



User Manual

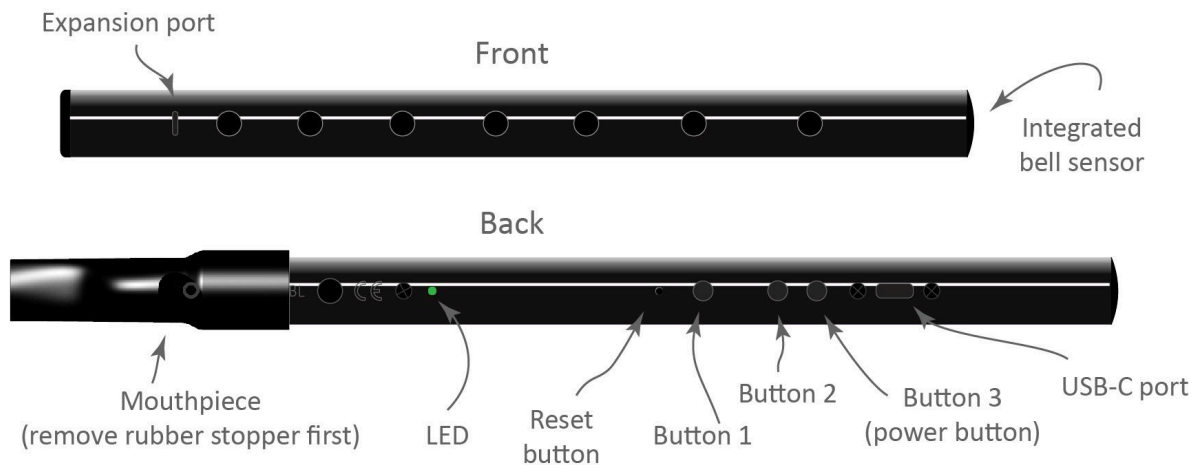
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1. Introduction

WARBL₂ is a USB-MIDI and BLE-MIDI wind controller, designed to simulate traditional open-tone-hole instruments like tin whistles, flutes, and bagpipes. WARBL₂ makes no sound by itself, instead sending MIDI messages to a MIDI module app or dedicated hardware MIDI module. This gives you complete and unlimited control over the sound.



WARBL₂ has nine optical tone-hole sensors, a pressure sensor, three buttons, and an LED indicator. The buttons are numbered 1, 2, and 3 from the top (mouthpiece) end of the device.

There is also a reset button under the small hole in the back of the enclosure. It can be clicked using a toothpick to reset the device (or power it down if under

battery power) and can be double-clicked to enter programming mode for updating the firmware.

WARBL₂ is highly customizable using the WARBL Configuration Tool, which is a web page accessible with a browser (most devices) or the WARBL app (iOS).

WARBL₂, introduced in 2024, includes a number of additions and improvements from the original WARBL. These include BLE MIDI, an inertial measurement unit (IMU: accelerometer and gyroscope), USB-C, an integrated bell sensor, an RGB LED, a faster processor with much greater storage space, a secondary processor dedicated to reading the optical sensors, higher-resolution pressure-sensor readings, lower power consumption, drag-and-drop firmware upgrades, slightly larger toneholes, and the ability to customize fingering charts completely.

WARBL₂ also has an expansion port for future upgrades or for use by end users who are proficient with electronics and coding. The WARBL₂ code contains detailed information about the expansion port.

2. Basic Operation

To power on WARBL₂ either click button 3 once or plug into USB power (a charging brick or a USB host device such as a computer, tablet, or phone). When plugged into a host device, WARBL₂ should appear as a USB MIDI device. To plug into a device that doesn't have a USB port (like an iPhone with a Lightning port) or to add a headphone jack to your device, you may need an adapter. Please see the [WARBL FAQ page](#) for adapter suggestions.

To power down WARBL₂ when under battery power, hold down button 3 for approximately 3 seconds (this is the default behavior but it can be changed using

the WARBL Configuration Tool—see section 7). By default the WARBL₂ will also power itself down after 5 minutes of inactivity (that can also be changed using the Configuration Tool).

To power down the WARBL₂ when powered by USB, simply unplug it.

Note: When storing the WARBL₂, ensure that the power button (button 3) isn't continuously pressed, for example by a tight-fitting storage container. This will cause the device to stay powered on and will drain the battery.

3. Battery information and charging

The rechargeable battery is a 1.2 V, 400 mAh $\frac{2}{3}$ AAA NiMH cell. The battery should allow for at least 10 hours of playing time per charge and should have a lifespan of at least a few years (likely several years). It can be replaced by the consumer at the end of its lifespan (see section 11). Replacement batteries are available in most countries or can be ordered from the WARBL website. The advantage of NiMH batteries over lithium ones is that they are safer (less flammable) and can easily be shipped overseas.

The battery voltage and charge percentage are shown in the Configuration Tool. There is also an option to program a button action that will cause the LED to flash a number of times corresponding to the charge percentage (see section 7).

To charge the WARBL₂, plug it into a good-quality USB charging brick or a USB host device as described above. The battery will begin charging if the battery voltage is lower than approximately 1.35 V. Charging is indicated by a pulsing red LED and there will be a charging icon next to the battery-level indicator in the Configuration Tool. The LED will turn solid green when charging is complete. A

fully depleted battery will take approximately 3 hours to charge fully. The WARBL₂ may get slightly warm while charging. It is okay to play the WARBL₂ while charging. A sophisticated charging algorithm is used to constantly monitor the battery voltage and temperature to ensure safe charging and prolong battery life.

Charging from a USB host can be turned off in the Configuration Tool (section 7), which is recommended if you are using USB MIDI and the WARBL₂ is often connected to a host. While charging, the WARBL₂ will draw 100 mA of current and may run down a phone battery. If this option is turned off, the WARBL₂ will only charge from a charging brick.

The WARBL will only begin charging if the battery voltage is lower than approximately 1.35 V. Other than that it can be charged at any time, but to calibrate the battery level monitor it is a good idea to occasionally use the WARBL₂ until the battery dies (when the battery voltage drops to 1.0 V the LED will illuminate red for 5 seconds and the WARBL will shut down) and also occasionally charge the WARBL₂ completely (until the LED turns green). It is also a good idea to fully charge the battery when you first receive the WARBL₂. The battery level is an estimate based on your playing habits, and it is not guaranteed to be accurate.

If you are not using the WARBL₂ often, it is a good idea to charge it every several months to help extend the life of the battery.

4. *Optical sensors*

The infrared optical sensors detect the distance of your fingers from the tone holes. The device uses this information to send MIDI pitch-bend messages, and to determine whether each hole is covered. The sensors may require occasional cleaning (see section 9) and calibration (see section 7). Although the device has been calibrated once when new, you will likely need to calibrate it to your own fingers.

Because they use infrared light, the sensors work well in most lighting conditions (including complete darkness). Outdoors in extremely bright sunlight, the effectiveness of the sensors may be impaired. If this occurs it may be necessary to turn away from the sun or play in a shady location. For best results, ensure that your hands are clean and dry. Oils on your hands may be invisible to you but still affect the reflectivity of your fingers in the infrared range. Recalibrating the sensors will fix most issues that arise, including if you are wearing gloves or have a bandage on a finger.

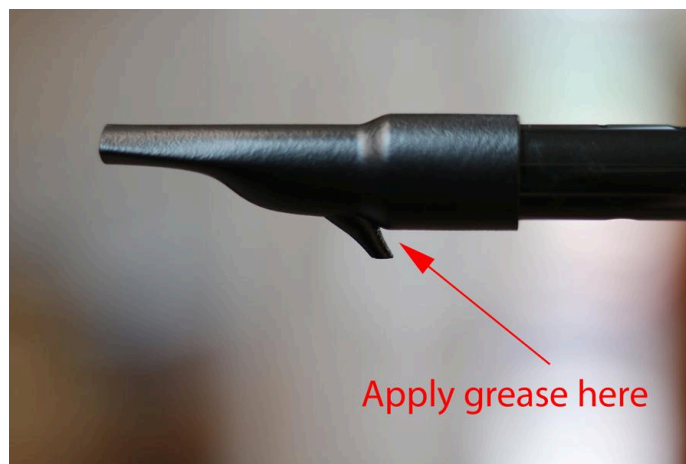
The integrated bell sensor is an additional infrared sensor designed to allow uilleann pipes players to play “on the knee”, stopping all sound when the bell sensor and all of the tone hole sensors are covered. This works with all fingering charts, not only the uilleann pipes chart. The bell sensor may also be used for vibrato.

5. *Pressure sensor and mouthpiece*

The pressure sensor measures the pressure from your breath or a bagpipe bag. The sensor is compensated for temperature and is quite precise. It can detect a maximum pressure of about 6 kPa (about 24 inches of water).

The mouthpiece is made of tough food-safe nylon, and is designed to direct air and moisture downward while allowing WARBL₂ to sense the breath pressure. To install it, first remove the rubber stopper from WARBL₂, and push and twist the mouthpiece in place (it doesn't need to be too tight). It has a tapered fit. To clean and sterilize the mouthpiece, you can use soapy water and/or boil it in water.

The mouthpiece exhausts excessive moisture out of the small “nozzle”. If you have any issues with moisture running down the back of the mouthpiece rather than dripping away, it helps to apply a small amount of silicone grease on the nozzle as shown below.



Please see section 8 for using a bagpipe bag instead of a mouthpiece.

6. MIDI module options and setup

The WARBL₂ doesn't produce any sound on its own and must be connected to a host device running a MIDI app that will produce the sound. Most mobile devices and computers can be used as MIDI host devices (either USB MIDI or BLE MIDI). IOS devices work especially well because they have relatively low audio latency and there are many low-cost MIDI app options. While many WARBL owners use Android devices, they are not generally recommended for MIDI because they can have higher latency.

For more advanced users, hardware MIDI Modules are another option for producing sound. The [EaganMatrix Micro](#) was designed to use with the WARBL and is a good option.

USB MIDI

The WARBL₂ should be recognized as a USB-MIDI device when plugged into a host device. Plugging WARBL₂ into mobile devices may require an adapter. Please see the WARBL [FAQ page](#) for up-to-date information on adapters for specific devices. Many new mobile devices have a USB-C port, in which case you can simply use a cable with USB-C plugs on both ends to connect the WARBL₂. However, if your device doesn't have a headphone jack and you'd like to add one, you may need an adapter or USB-C hub to do that.

BLE MIDI

WARBL₂ can communicate wirelessly with many host devices using BLE MIDI. BLE MIDI is a short-range protocol and the maximum range is approximately 30

feet (roughly 10 meters). At longer distances latency may increase and notes may begin to drop out.

iOS Devices

On **iOS devices**, the WARBL₂ should reconnect automatically each time it is turned on. To pair the device the first time, you can use the WARBL app or any other MIDI app that includes the ability to pair BLE devices [Korg BLE-MIDI Connect](#) or [MIDI Wrench](#) (both free).

To pair the WARBL₂ in the WARBL app (shown below), click on “BLE MIDI” at the top. The WARBL₂ should be visible in the list of devices if it is turned on and within range. Click on the “WARBL” listing and then click “pair” when prompted.



Once paired, the WARBL₂ should work with any MIDI app, for example the WARBL app or Celtic Sounds MIDI Module. You should be able to simply turn on the WARBL₂, open a MIDI app, and begin playing. You should only have to pair the WARBL₂ again if it becomes “forgotten”. If you have any trouble reconnecting the WARBL₂ you can “forget” it in your device’s settings and then pair it again.

Windows machines

Not all **Windows** machines have built-in support for BLE devices, but most newer ones do. If yours doesn't you can buy a BLE dongle to add that functionality. On Windows, a helper app is currently required to make BLE-MIDI devices appear seamlessly as normal MIDI devices. Microsoft is currently working on making this more seamless, but it may be a year or so before that is complete. In the meantime, I highly recommend the PerfectBluetoothMIDI app. To use it, download the latest "perfectBluetoothMidi.exe" file [from this page](#). You can put this anywhere on your machine, but if you'd like it to run automatically at startup (recommended), put it in your Windows startup folder. Then double-click it to run it manually the first time. Next, turn the WARBL₂ on, make sure Bluetooth is turned on on your machine. There should be a device shown in the "Bluetooth MIDI Device" list. Select that and click "Connect". The WARBL₂ should connect and then reconnect automatically each time you turn it on. If you have the app in your startup folder and you don't want it to be visible after startup, you can select the option in the settings to "automatically minimise to the system tray".

Another great option for connecting to Windows is to use a [CME WIDI Bud Pro](#) adapter. This plugs into a USB port, connects reliably to the WARBL₂, and shows up as a USB MIDI device on your computer.

Note: On older versions of **Windows**, only one app is able to connect to a MIDI device at a time. So, if you've been connected to the Configuration Tool you'll need to close your web browser before you can connect to a MIDI app, and vice versa.

Mac OS

To connect the WARBL₂ to a **Mac**, you can follow the directions [in this link](#) to “Make your Mac a Bluetooth Host”.

Android

MIDI apps on **Android** like FluidSynth and BS-16i should be able to connect directly to the WARBL₂, but if this doesn’t work it’s possible that a helper app like [MIDI Connector](#) or [BLE Connect](#) will help.

If you experience issues with BLE MIDI not connecting correctly on any device, a solution is often to have your device “forget” the WARBL₂ and then reconnect to it.

Important: The WARBL₂ is capable of connecting by both USB MIDI and BLE MIDI simultaneously. While this can be useful in some cases (e.g. sending messages to two different devices), it can also be a cause for confusion. You can choose in the Configuration Tool whether to send MIDI notes to either USB or BLE, or you may simply wish to turn off BLE on your devices if you are using USB MIDI with the WARBL₂. In general, it is not a good idea to be sending both BLE and USB MIDI messages simultaneously to a single device.

BLE latency

BLE MIDI adds a small amount of latency (delay). On most devices this is imperceptible or nearly so, but for the absolute lowest latency you may choose to use USB MIDI instead. On newer devices, the average added BLE latency is

usually around 4-8 milliseconds (a BLE connection interval of 7.5 - 15 ms). The BLE connection interval is shown in the Configuration Tool.

For comparison, 8 milliseconds of latency is the amount of time needed for sound to travel 8 feet, so it is the equivalent of standing 8 feet away from an audio source. In fact, for keeping perceived latency to an absolute minimum it is recommended to be close to your audio source (speakers) because just standing several feet from speakers will effectively double the overall latency.

Having the WARBL₂ connected to the Configuration Tool may also add a very small amount of latency because it necessitates sending additional data wirelessly. You can click “Disconnect” in the Configuration Tool when done using it to avoid this.

Note that there are many other sources of latency. If you are experiencing a large amount of latency, it's very likely the result of audio latency associated with your host device, rather than BLE. A few things to check are:

- Are you using Bluetooth headphones or speakers? These always add a large amount of latency and can't be used with live music performance (Bluetooth audio is much slower than BLE MIDI). You'll need to use wired headphones or speakers.
- Are you using a Windows machine? If so, try selecting different audio drivers in your MIDI app settings, and possibly install the free ASIO4All audio driver.
- Are you using an Android device? Some of these have a lot of audio latency and there may not be a solution for this. I'm happy to accept returns if you are unable to attain acceptable results with your Android device.

The [CME WIDI Bud Pro](#) adapter is a great way to ensure that the WARBL₂ connects with a 7.5 ms interval (the fastest currently possible).

MIDI Apps

The WARBL Configuration Tool now has built-in audio with several instrument sounds available. However, this is mostly meant for testing purposes or informal playing, and it's generally better to use a dedicated MIDI app or synth.

Some recommended iOS MIDI module apps are Appcordians [Celtic Sounds MIDI Module](#) (or other Appcordians MIDI apps), Thumbjam, and SoundFonts. WARBL₂ normally sends melody (chanter) notes on MIDI channel 1, which most apps will listen to by default. If the app allows control over the maximum pitch bend, it should be set to two semitones. This is the default for many apps. Some apps allow “background play”, meaning that the app will continue to play after it is closed. This allows using multiple apps simultaneously, producing the sound of multiple instruments at once. It also allows you to use the WARBL Configuration Tool simultaneously with a MIDI module app.

For platforms other than iOS, there are two new Web pages by Michael Eskin that connect to the WARBL (much like the Configuration Tool does) and supply high-quality audio in a web browser like Chrome that supports Web MIDI. They are [Celtic Sounds \(web version\)](#) and the [WARBL Sound Lab](#).

For Android devices, the [BS-16i](#) or [FluidSynth](#) apps are recommended (BS-16i has more features and a better user interface). On Windows and Mac, [SForzando](#) can work well. These apps all play SoundFonts and you can download the [WARBL SoundFont](#) for playing whistle, flute, uilleann pipes, highland pipes, and smallpipes sounds. For bagpipes sounds on Windows and

Mac, [Universal Piper](#) is excellent. There are many other free SoundFonts on the web if there are other instrument sounds you'd like to make with the WARBL₂.

Note: MIDI apps that are open “in the background” continue to use battery life, so it's a good idea to force them to close after you are done playing ([here's how to force-close an app on iOS](#)).

Hardware Synths

The [EaganMatrix Micro](#) sound engine is a compact hardware synth that was designed specifically for use with the WARBL. The web site has a demo video and information about how to use it. It is great for more advanced/demanding users and has exceptional sound quality.

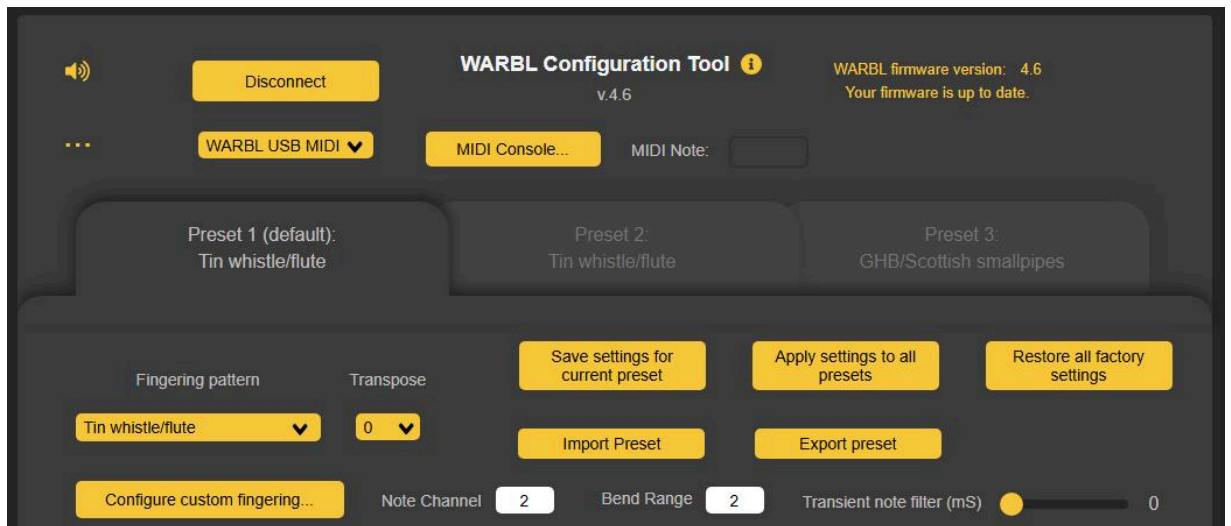
Most other MIDI hardware synths should also work with the WARBL₂.

On the WARBL Forum there is a growing list of presets available to help set the WARBL up with appropriate settings for various apps and synths.

7. Customization with the WARBL Configuration Tool

Introduction

The Configuration Tool is a web page that connects to WARBL₂ using MIDI protocol to let you customize settings. This section is organized to correspond with the panels in the Configuration Tool, and most of the information in this section is also available in the Configuration Tool by clicking on the yellow “info” icons.



Because the Configuration Tool uses the relatively new Web MIDI protocol, It is necessary to use a browser that supports Web MIDI, which currently includes Chrome, Firefox, and Opera on Windows, Mac, Linux, Chromebook, or Android.

Safari unfortunately does not yet support Web MIDI, but on iOS devices you can use the free [WARBL app](#), generously made by Michael Eskin of Appcordions.

The Configuration Tool will try to connect to WARBL₂ if it is already plugged in or connected by BLE MIDI when the page is loaded, or you can click **Connect to WARBL** if it hasn't connected automatically. Once it has connected, it should show your firmware version in the upper right corner, and it will tell you if there is a firmware update available (see section 10 for updating the firmware). **NOTE:** If your firmware is not up to date, you will be able to use the Configuration Tool normally, but some options may be missing or disabled (shown in gray).

Top Controls

WARBL₂ has three "**presets**" available for use at any time. Each preset consists of a fingering pattern and a set of associated settings. You can select different fingering patterns for each preset, or you can select the same fingering pattern for more than one preset. For example, if you only want to use tin whistle fingering, you can select that option for all three presets, but still have three different sets of settings available at any time. You may want to have the buttons configured differently for each preset, for example, or have different register control and/or slide/vibrato settings.

Selecting the tab for each preset allows you to change the settings for that preset. **Please note** that WARBL₂ responds immediately to changes to settings for each preset, but if you don't save them, those changes will be lost when you power down WARBL₂. You can choose to apply changes to settings to the current preset (by clicking **Save settings for current preset**), or you can choose to apply them to all three presets (click **Apply to all presets**). This can be useful if you want to make three presets that are the same except for minor differences (which can be changed after applying the initial settings to all three presets). You

can also **restore the factory settings**. This will put the WARBL₂ in the same state as when you received it, including the factory sensor calibration.

You can set the **Default preset** that will be active when WARBL is plugged in. This prevents having to click a button to switch presets if your current favorite preset is not preset 1. To set the default preset, simply click the tab for that preset and then click **Set as Default**. This change will be saved immediately. **For advanced users:** To tell the WARBL to switch between presets, you can send Program Change (PC) messages 1, 2, or 3 on any MIDI channel from another device.

The **Preset Import and Export** buttons allow you to exchange your WARBL preset settings with other users or to archive settings for one preset for later import back into the WARBL Configuration Tool. On the WARBL Forum there is a growing list of presets available to help set the WARBL up with appropriate settings for various apps and synths.

WARBL settings for the currently selected preset may be exported to a file using the **Export Preset** button.

The exported file will be named "WARBL_Preset.warbl" and can be found in the "Downloads" directory on your computer or typically in the Files app in iOS. If there is already an existing export file in that directory, Chrome will save the file with incrementing numbers after the "WARBL_Preset" name.

To load an existing WARBL preset file into the current preset tab, click the **Import Preset** file selector button.

Select a .warbl settings file from your computer to import. The imported data will be loaded into your WARBL, but not saved permanently to allow you to make additional changes if desired.

To permanently save the settings for the current preset in your WARBL, click the "**Save settings for the current preset**" button at the top of the screen.

Note: It is not possible to import settings from an original WARBL to a WARBL2 or vice versa because they use different commands for communicating with the Configuration Tool.

The **MIDI note** display will show what MIDI note is currently being played, and the **MIDI Console** will show the most recent 300 MIDI messages that have been received. These are both useful for troubleshooting.

The **Fingering pattern** dropdown menu allows you to choose a fingering chart. Information about each chart is available on the [Documentation](#) web page. Note that there are four **Custom** charts available at the bottom of the list (see Custom Fingering Configuration below).

The **Transpose** menu allows you to change keys. If this is set to 0 (the default) then the selected fingering chart will produce the notes shown in the documentation. You can transpose up or down as much as 18 semitones to play in different keys. Most MIDI apps also allow transposing the received MIDI notes, and often it is simpler to do this in the MIDI app rather than in the Configuration Tool.

The **Note channel** input allows you to select the MIDI channel (1-16) used for sending notes from WARBL. This should typically be left set to "1".

The **Bend Range** setting is the number of semitones of pitch bend that the WARBL will allow. This should typically be set to the default value of 2.

The **Transient note filter** filters out very brief unwanted notes by adding a small delay to the tonehole sensors. This can be particularly useful when using note

annotation software to transcribe your playing. Without the filter, there can be very brief notes that you might not necessarily hear but that will be transcribed. The filter can also be used to "clean up" the sound of your playing by removing "crossing noises" that occur while changing from one fingering pattern to another. Be aware, however, that because the filter works by adding a delay, setting it too high will give a "sluggish" feel. Set it to "0" for the fastest possible tonehole response.

Audio player

The Configuration Tool now contains an audio player with several instrument sounds available. To turn on the audio, click the  button in the top left corner of the Configuration Tool. To select the instrument sound and other settings, click the "... " button directly below.

The audio is mostly for testing purposes, and a standalone app is generally recommended for the best sound quality. Note: on iOS devices, you might not hear the audio if the **Do Not Disturb** switch is turned on. If you are not hearing sound, use the **Test Audio** button to determine if the problem is with the audio output or your WARBL settings.

The audio will respond to **CC2, CC7, CC11, or channel pressure messages** for controlling loudness. If no expression messages are received for two seconds while a note is playing, loudness will be reset to the maximum.

The **Transpose (semitones)** can be used to change the key of the audio. If you are using bagpipes sounds, use this rather than the other Transpose selector next to the Fingering Pattern menu, because this will keep the drones in tune with the chanter. This **only** affects the audio output here, **not** the MIDI notes that

the WARBL is sending. **Note:** Transposing the uilleann pipes sound lower than -7 semitones is not recommended because it then extends into the drones range and low notes will cause the drones to play.

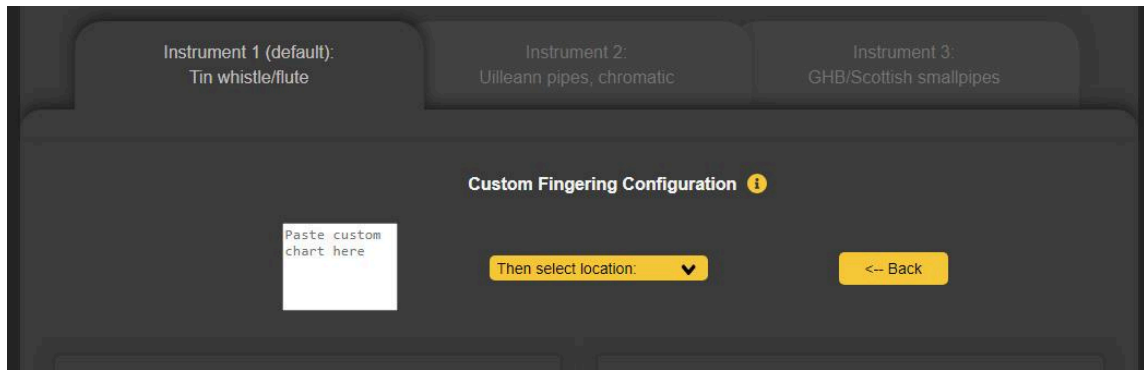
The bagpipes contain drone sounds that can be turned on by using the **Drones** panel to a MIDI **Note On** message (any note lower than 55). They are turned off by sending a **Note Off** message for the same note number, or a **Note On** message with a velocity of 0 (byte 3 in the drones panel). To ensure that the drones don't respond to pitchbend messages, you can use a different channel for drones than the main note (chanter) channel.

The MIDI pitchbend range in the audio player is automatically synced with the **Bend Range** in the current WARBL preset.

To reset the audio, use the  button to turn it off and then on again.

Custom Fingering Configuration

WARBL₂ has the ability to store up to four completely customizable fingering charts. Each chart contains MIDI notes for every possible combination of the 8 tone holes (the bell sensor is not used). There are 256 such possible combinations. To upload a custom fingering chart to the WARBL₂, click **Configure custom fingering....** The screen below is displayed.



You will be pasting a list of 256 MIDI notes into the field at the left. To create your custom fingering chart, there is a special Google Sheet available. [Follow this link](#) to access the sheet (shown below). You don't have to use this sheet but it helps simplify the process of making a chart. **You'll need to save your own copy of the sheet in order to edit it.**

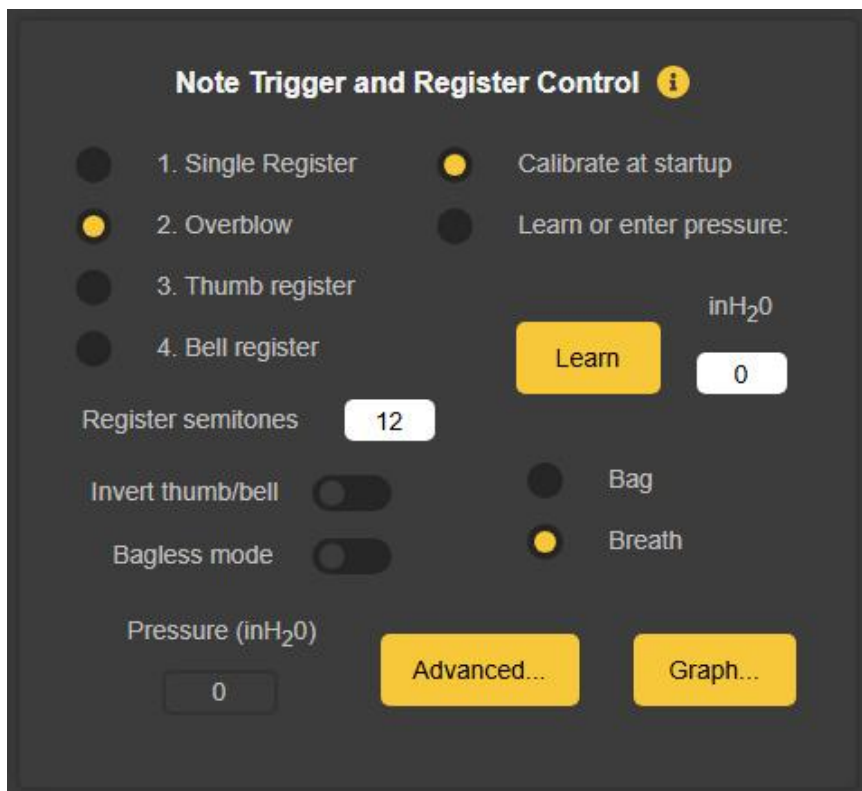
	A	B	C	G	H	I	J	K	L	M	N	O
1	Fingering pattern (0 is an uncovered tone hole, 1 is a covered hole)	Note name (optional)	MIDI Note (fill this column in with your desired MIDI note for every fingering pattern, and then copy/paste into the Configuration Tool).	Sample charts (You can start with one of these, copy it into column C, and modify it as necessary):	Tin whistle/flute, chromatic	Tin whistle/flute, standard	Uilleann, chromatic	Uilleann, standard	GHB/Scottish smallpipes	Northumbrian smallpipes	Gaita	Gaita, ex
2	0 000 0000				73	73	74	74	82	77	72	
3	0 000 0001				73	72	74	74	82	77	72	
4	0 000 0010				73	73	74	74	82	77	72	
5	0 000 0011				73	72	74	74	82	77	72	
6	0 000 0100				73	73	74	74	82	77	72	
7	0 000 0101				73	72	74	74	82	77	72	
8	0 000 0110				73	73	74	74	82	77	72	
9	0 000 0111				73	72	74	74	82	77	72	
10	0 000 1000				73	73	74	74	82	77	72	
11	0 000 1001				73	72	74	74	82	77	72	
12	0 000 1010				73	73	74	74	82	77	72	
13	0 000 1011				73	72	74	74	82	77	72	
14	0 000 1100				73	73	74	74	82	77	72	
15	0 000 1101				73	72	74	74	82	77	72	
16	0 000 1110				73	73	74	74	82	77	72	
17	0 000 1111				--	--	--	--	--	--	--	

Column A lists all possible fingering combinations (0 represents an open hole and 1 represents a closed hole). You will need to enter a MIDI note number (0-127) in column C for each fingering combination. This is less tedious than it sounds because many will be duplicates— you can enter MIDI notes for the “normal” fingerings and then autofill empty cells below each normal fingering. Notice that for reference the MIDI notes for all the “standard” fingering charts are shown starting in column H. You can choose to modify an existing chart by copying the values from it into column C and then editing it as desired.

If you'd like you can save a copy of the Google sheet for your own use. It may be a good idea to do this or copy your custom MIDI values into another document so you have a record of them.

Once you have filled in all 256 in column C, select all of those values and copy them. Returning to the Configuration Tool, paste the values into the field at the left. Next, select one of four custom charts in the center dropdown menu as the location for saving a chart (if you have previously saved a chart in this same location it will be overwritten). It will take several seconds to load the new chart onto the WARBL₂. After you have done this, click **Back** and then select your custom chart from the Fingering Pattern dropdown menu (and save changes if desired).

Note Trigger and Register Control



The screenshot shows a configuration window titled "Note Trigger and Register Control" with an information icon. It contains several controls for setting up note triggering and register control.

- Register Selection:** Four radio buttons are listed: "1. Single Register", "2. Overflow" (selected), "3. Thumb register", and "4. Bell register".
- Calibration:** A radio button for "Calibrate at startup" is selected.
- Pressure Entry:** A section labeled "Learn or enter pressure:" includes a yellow "Learn" button and a numeric input field showing "0" with the unit "inH₂O".
- Register Semitones:** A numeric input field is set to "12".
- Invert thumb/bell:** A toggle switch is currently turned off.
- Bagless mode:** A toggle switch is currently turned off.
- Pressure (inH₂O):** A numeric input field at the bottom left shows "0".
- Mode Selection:** Two radio buttons are present: "Bag" (selected) and "Breath".
- Action Buttons:** Two yellow buttons at the bottom right are labeled "Advanced..." and "Graph...".

You can play in a single register or choose from several options for controlling the register. Choosing **1. Single register** will allow you to use breath or bag pressure to start/stop notes, but will only allow access to one register. The pressure sensor can be used to mimic the way that whistles, flutes, and some bagpipes can be "overblown" to reach the second octave. To use this feature, select **2. Overblow**. For fingering patterns that don't use the left thumb, you can instead select **3. Thumb register** to allow you to use that finger to control the register, which will be familiar to players of many woodwind instruments. To play the upper register, simply uncover the thumb hole. **Note:** With the **Thumb register** option selected (with or without using the thumb for half holes), if you are using a custom fingering chart, the thumb position in your chart will be ignored and only the first half of the chart will be used. You do need to have values in the second half of the chart (the part with the thumb hole covered), but those values will be ignored.

Finally, you can use the bell sensor to select the register by choosing **4. Bell register**. Uncovering the bell sensor (lifting it off your knee), will access the second register. The functionality for either the thumb register or bell register options can be reversed by turning on the **Invert thumb/bell** switch. Then covering the left thumb hole or covering the bell sensor will allow playing in the second register, depending on whether option 3 or 4 is selected. In the case of the thumb hole, this will be more familiar to players of instruments with a register key.

Starting with firmware v. 4.4, you can select the number of semitones in a register. The default is 12, which means that all three above methods of controlling the register will change the pitch by one octave. You can change this to any number of semitones (up to 24), which is useful for emulating wind instruments like the kaval that naturally overblow at an interval less than an

octave. It also can serve other purposes, like playing accidentals using the above methods.

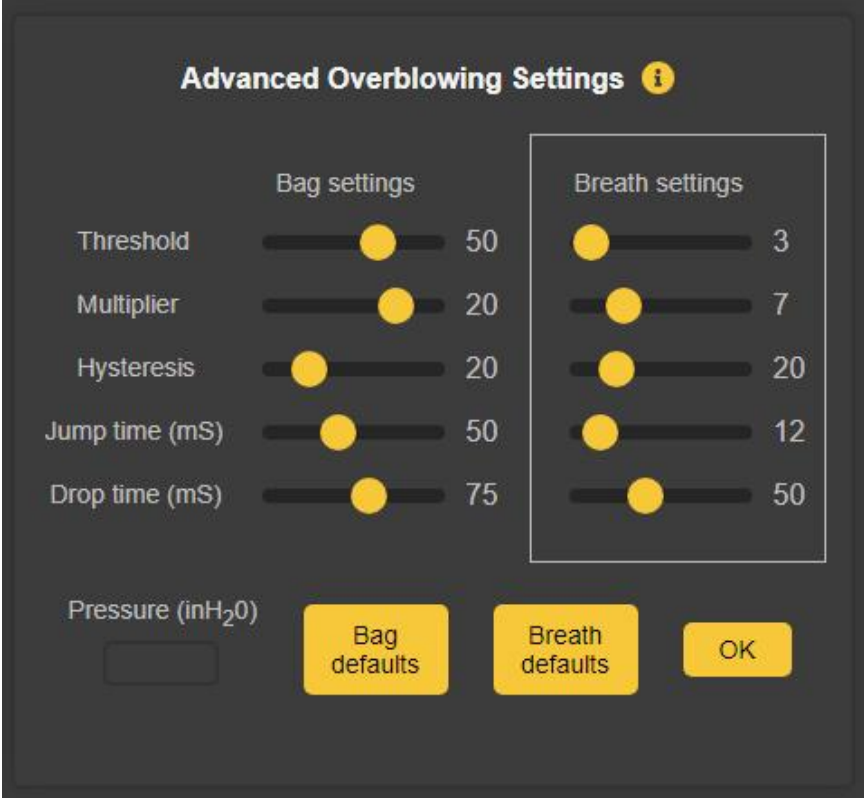
Note you can also use the IMU (elevation angle) to control the register. This is discussed in the Pressure and IMU Mapping section.

Turning on **Bagless mode** will allow you to configure a button to start/stop the sound instead of using the pressure sensor. This will be similar to many electronic bagpipe chanters that don't use pressure sensing. If you choose you can still select one of the options for controlling the register when using bagless mode. **Note:** if you choose the bagless option you must also configure a button to **Play/stop (bagless mode)**.

The pressure sensor will normally be calibrated at startup, meaning that the pressure when you power on the device will be used as a basepoint for determining the pressure level at which sound will be triggered. This is desirable if you are using breath to control the instrument. If you are using a bag with this setting, you will want to squeeze the bag when you power on WARBL₂ to "tell" it the pressure at which you want notes to begin playing. Alternatively, you can select **Learn or enter pressure** to use a different pressure. This is usually preferable if you are using a bag instead of breath, and you want to trigger notes at a higher pressure. You can squeeze the bag at the pressure that you would like to use for initial triggering of sound and then click **Learn**, which will cause WARBL₂ to use the current pressure (as with all settings, you can save this pressure by saving the settings for the current preset). After clicking "Learn", the pressure input cell will flash green and the learned pressure will be displayed. You can also enter the desired pressure manually if you prefer. Simply type in a pressure between 0.0 and 24.0 (inches water) and hit return or enter. The input cell will momentarily turn green to indicate that the pressure setting was sent to WARBL₂.

Selecting **Bag** will cause the overblowing settings for bagpipe bags to be used, while selecting **Breath** will cause the settings for breath (mouthpiece) to be used. These settings are customizable using the **Advanced** panel if desired (please see the help file within the Advanced panel for information about these settings).

Advanced Overblowing Settings



The image shows a software dialog box titled "Advanced Overblowing Settings" with an information icon. It contains two columns of settings: "Bag settings" and "Breath settings". Each column has five sliders with numerical values. At the bottom, there is a "Pressure (inH₂O)" input field, two buttons labeled "Bag defaults" and "Breath defaults", and an "OK" button.

Setting	Bag settings	Breath settings
Threshold	50	3
Multiplier	20	7
Hysteresis	20	20
Jump time (mS)	50	12
Drop time (mS)	75	50

Pressure (inH₂O)

Bag defaults Breath defaults OK

These settings control the way that the register is changed by overblowing. Each variable below has a setting for a **bag** and one for the **breath** (mouthpiece). which settings are used depends on whether you currently have **Bag** or **Breath**

selected in the main Note Trigger and Register Control panel (the currently selected group of settings are surrounded by a white box). The variables that are most likely to need adjustment are located near the top. Clicking the **Bag defaults** button will restore the default bag settings for all variables in the panel, and clicking the **Breath defaults** button will restore the default settings for breath. Units are not shown for some of these settings because they use arbitrary units.

Threshold represents how much pressure is required to move from the first register to the second register. Increasing this setting gives you more control over the registers but also makes it require more breath or bag pressure to overblow.

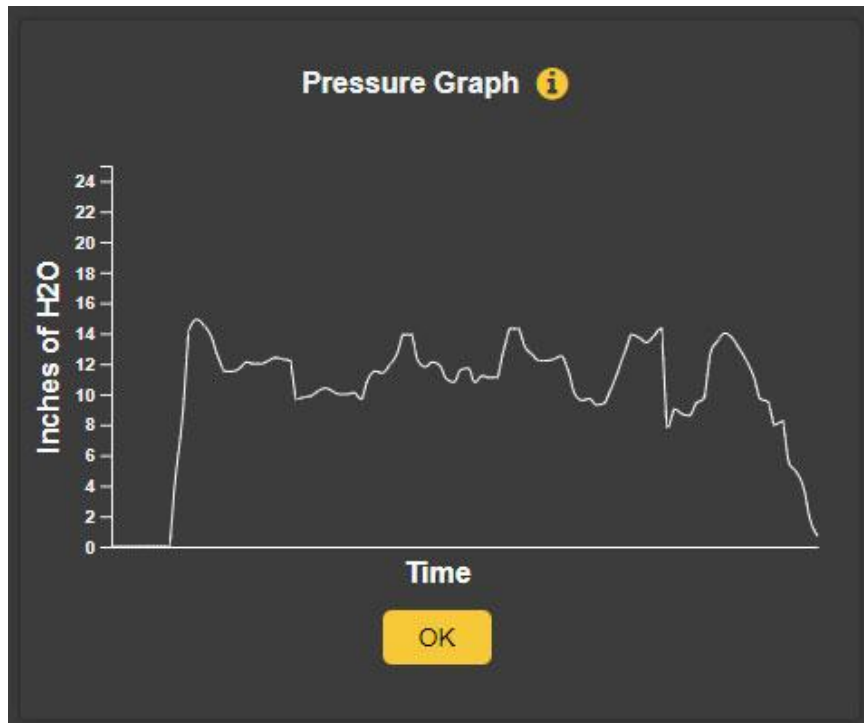
With wind instruments that respond to overblowing, it typically requires more pressure to move to the second register from higher notes in the scale than it does from lower notes in the scale. The **Multiplier** setting controls how pronounced this difference is. Again, increasing this setting can give you better control over the register, especially if using a bag, but will also increase the pressure required to reach the second register, from higher notes in the scale in particular.

Note: Setting both **Threshold** and **Multiplier** too high can make it impossible to reach the highest notes in the second register because the pressure sensor is saturated, i.e. it has reached the highest pressure it can detect. If you find that you cannot reach the highest notes in the second register, try reducing one or both of these settings slightly.

Hysteresis causes it to be slightly easier to stay in the second register than it is to initially reach that register. In other words, the pressure threshold for moving up to the second register is slightly higher than the threshold for moving back down. This is how real wind instruments behave.

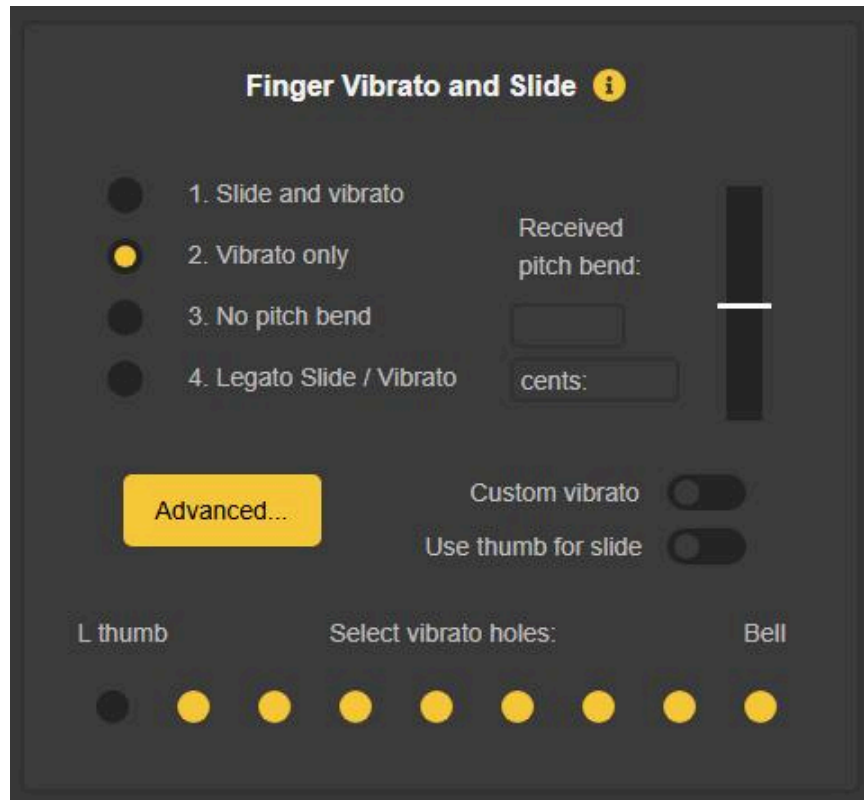
WARBL₂ has the ability to **"jump"** directly from silence to the second register or to **"drop"** directly from the second register to silence. Thus, by blowing forcefully (particularly by "tonguing" a note), you can begin playing in the second register without first playing a brief note in the first register. Similarly, by cutting off air pressure suddenly, you can drop directly from the second register to silence. WARBL does this by waiting a small amount of time for a register threshold to be crossed or for the pressure to level off, to detect the player's intent. Increasing the **Jump time** setting makes it easier to move directly to the second register by increasing the amount of time that WARBL waits to determine the intent of the user. Increasing the **Drop time** has a similar effect for moving directly from the second register to silence.

Pressure Graph



The pressure graph can be used to view the WARBL pressure sensor readings in inches of H₂O over time.

Finger Vibrato and Slide



WARBL₂ mimics open-tone hole instruments by sensing the distance of your fingers from the holes and using this information to bend notes downward. It can detect your fingers a maximum of about 1 cm (~1/2") from the holes.

You can choose **1. Slide and vibrato**, **2. Vibrato only**, or **3. No pitch bend**, or **4. Legato slide/vibrato**. The way **slide** works is that lowering the finger over the highest uncovered hole will gradually flatten the current note down to the next lower note on the scale, much like on a real instrument. This can also allow you to approximate "half-holing" accidentals. Because of the nature of note transitions in MIDI, sliding between notes can sometimes result in a slight "popping" sound when the new note is triggered. This is quite apparent with some sounds like whistles and less so with other sounds like reed instruments. Whether you choose to use slide may depend partly on the MIDI host and

sounds that you're using. The normal slide feature is limited to a maximum of 2 semitones before a different MIDI note is triggered.

The **4. Legato slide/vibrato** feature is different from the normal slide in that it allows you to slide seamlessly over a range of more than 2 semitones without triggering a new MIDI note. The sliding range is determined by the **MIDI bend range (semitones)** setting at the top of the Configuration Tool and the **Legato slide limit (semitones)** setting in the **Advanced Pitch Bend Settings** panel.

For proper slide and vibrato results, the pitch bend range in the MIDI app or host that you are using **must be set to the same number of semitones** as the **MIDI bend range (semitones)** setting. The Legato slide/vibrato feature also uses additional holes as “slide holes”: Rather than using only the highest uncovered hole, it uses any hole that would normally cause a new note to be triggered if it was covered.

With **vibrato**, lowering fingers over any open holes (other than the "slide" hole, if you also have slide turned on) will gradually flatten the note down to the selected maximum vibrato depth. The pitch bend contributed by each hole is cumulative, meaning that lowering fingers over multiple holes simultaneously will flatten the note more than lowering just one finger. This gives control over the amount of vibrato, and can be used to simulate sliding even when "slide" isn't turned on. If a finger is covering a hole when a note is triggered and subsequently lifting that finger doesn't cause a different note to be triggered, then that finger will be available to use for vibrato. However, to enable that finger, you first must remove it completely, past where WARBL can first detect it. This is to prevent problems with closed fingering patterns. As an example of how this works, if you trigger a note, completely remove a finger (one that doesn't change the current note), and then place it back down, the resulting pitch will be lower than the originally triggered note because vibrato has been enabled. This is usually intuitive, but it's

important to understand because in certain cases it might not be immediately apparent why the resulting pitch is lower than the original pitch.

You can select at the bottom of the panel which holes you'd like to have available for vibrato. If you normally only use a few fingers for vibrato, you may want to select only those holes, to prevent unwanted behavior.

Custom vibrato is a different vibrato option that is only available for some fingering patterns (currently tin whistle, uilleann pipes, GHB, and Northumbrian pipes). For **tin whistle and uilleann pipes**, the differences from normal vibrato are as follows: 1. Only the first and/or second fingers of the right hand are available for vibrato, depending on the fingering pattern (you cannot select vibrato holes if this option is turned on). 2. Vibrato is not cumulative, meaning that lowering more than one finger doesn't increase the vibrato depth. 3. If a vibrato hole is covered when a note is triggered, the note is immediately flattened (this is the major difference and may be more natural in certain circumstances). 4. When playing a back D with uilleann fingering, uncovering a vibrato hole actually lowers the pitch instead of raising it, which is how real pipes behave. For **GHB and Northumbrian pipes**, the custom vibrato is designed for closed fingering systems, and sharpens the note when finger R2 or R3 is raised from a closed fingering position (as opposed to the normal vibrato, where lowering a finger from an open position flattens a note).

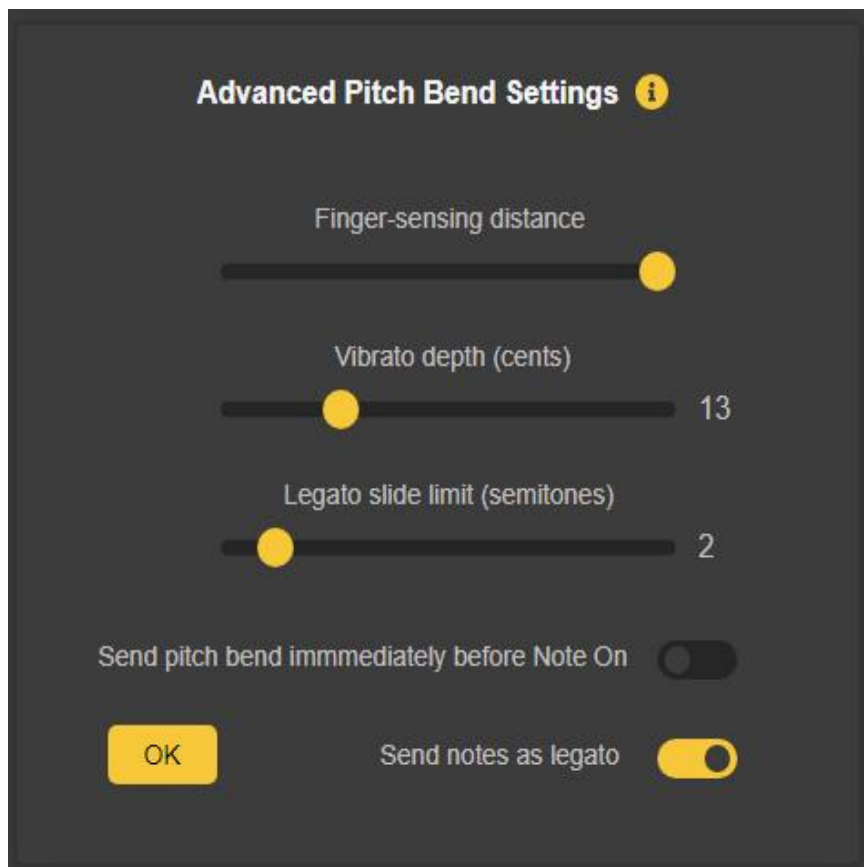
Note that custom vibrato is not available when any unsupported fingering pattern has been chosen.

If you'd like to prevent the thumb hole from being used for slide, you can turn off the **Use thumb for slide** switch.

The pitch bend value received from the WARBL is shown at the right, along with a "wheel"-type display. Note that any received pitch bend messages will be

displayed here, including those resulting from IMU vibrato or pitch expression, which are controlled in other panels.

Advanced Pitch Bend Settings



Accessed by the “**Advanced**” button in the **Finger Vibrato and Slide** panel, these settings can be left unchanged by many users, but give more control over how WARBL sends pitch bend messages.

The **finger-sensing distance** is the relative distance at which the toneholes sense your fingers for slide and vibrato. You can reduce this distance using the slider if you don't want to have to remove your fingers as far from the holes to turn off the pitch bend. The downside is that you'll have less precision in bending notes.

The **Vibrato depth (cents)** slider determines the degree to which notes are bent when using finger vibrato.

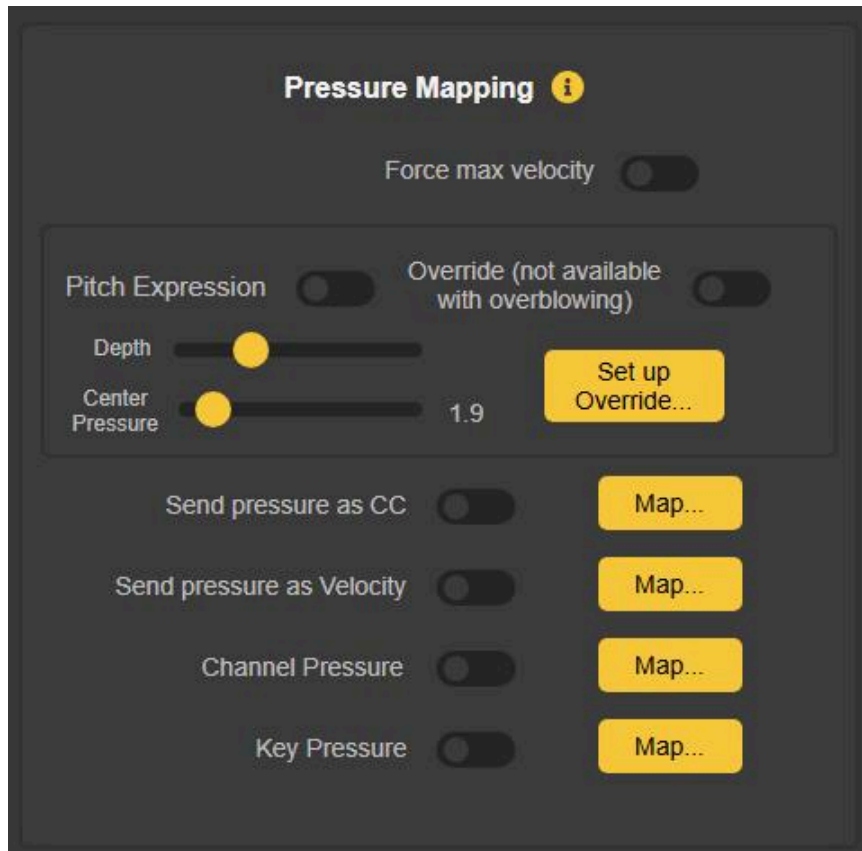
The **Legato slide limit (semitones)** setting determines the maximum range for sliding if you are using the **4. Legato slide/vibrato** option. Note that the **MIDI bend range (semitones)** must also be set at least as high, and you must also always set the MIDI pitch bend range in whatever MIDI app or host you are using to the same value as the **MIDI bend range (semitones)** setting.

Normally WARBL₂ sends a pitch bend message after a new MIDI note is turned on. Some users may instead wish to send pitch bend before the note is turned on by using the **Send pitch bend immediately before Note On** switch. This may cause pitch bend audio artifacts with some MIDI apps but is the recommended setting for MIDI MPE.

The **Send notes as legato** switch cause WARBL₂ to wait to turn off each MIDI note until after the next note is sent. This is the recommended setting for most apps and allows many apps to interpret the notes as legato, allowing them to flow seamlessly together.

Pressure Mapping

This panel allows mapping of breath (or bagpipe bag) pressure to various MIDI messages.



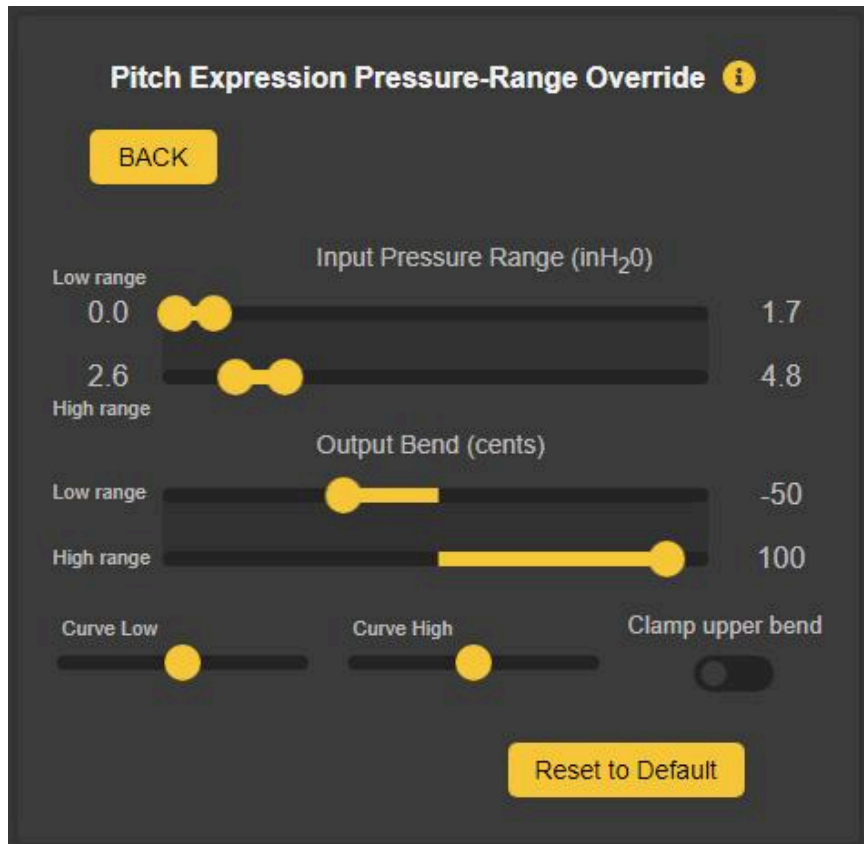
Turning on the **Force Max Velocity** switch will cause WARBL₂ to always send notes with a velocity of 127 (the maximum allowed), which will allow maximum volume from MIDI apps. Some more sophisticated apps interpret high velocity in ways that also affect the timbre of notes, or various other effects. If this is the case, you may want to uncheck this switch, which will cause WARBL₂ to send notes with a velocity of 64 (the middle of the allowed range). The switch has no effect if the **Send Pressure as Velocity** switch (below) is selected.

Turning on the **Pitch expression** switch will cause WARBL to send pitch bend messages based on pressure. This means that for a given note, blowing (or squeezing a bag) gently will cause the note to play slightly flat, while blowing with excess force will cause it to be slightly sharp. This is meant to simulate the way

that pressure is used to fine-tune the pitch of real instruments, and can make WARBL₂ sound more lifelike. The **Expression depth** slider controls the amount of pitch bend, and on firmware 4.3 and above the **Center Pressure** control lets you set the preferred blowing pressure where the pitch is centered. The pitch bend is added to any that is contributed by vibrato and slide, so they can all be used simultaneously. If overblowing is not being used, you have the option of changing the default pressure range used for Expression by clicking the **Override...** button. On firmware 4.3 and above, the pitch expression override controls are much more advanced, letting you control exactly how the bend responds to pressure.

WARBL₂ also has the ability to map pressure sensor values to a MIDI Continuous Controller (CC) note-on velocity, channel pressure, and key pressure, allowing it to function as a MIDI breath controller. These are advanced settings and not needed by many WARBL users. An example would be to send pressure data to CC 7 to control the volume of notes with pressure. The **Send Pressure as Velocity** option overrides the **Force Max Velocity** switch (above).

Pitch Expression Pressure Range Override



The image shows a settings screen titled "Pitch Expression Pressure-Range Override" with an information icon. It features a "BACK" button and several sliders for configuring pitch expression. The "Input Pressure Range (inH₂O)" section has two sliders: "Low range" (0.0 to 1.7) and "High range" (2.6 to 4.8). The "Output Bend (cents)" section has two sliders: "Low range" (-50 to 0) and "High range" (0 to 100). At the bottom, there are sliders for "Curve Low" and "Curve High", a "Clamp upper bend" toggle switch, and a "Reset to Default" button.

Section	Parameter	Value 1	Value 2
Input Pressure Range (inH ₂ O)	Low range	0.0	1.7
	High range	2.6	4.8
Output Bend (cents)	Low range	-50	0
	High range	0	100

By turning on the **Override pitch expression range** switch and using the sliders, you can override the default pressure range used for pitch expression. This is not available if overblowing is being used, because in that case the pressure range has to be calculated in real time based on the current register and overblowing settings.

With the original WARBL, there is a single slider for controlling the pressure range used for pitch bend.

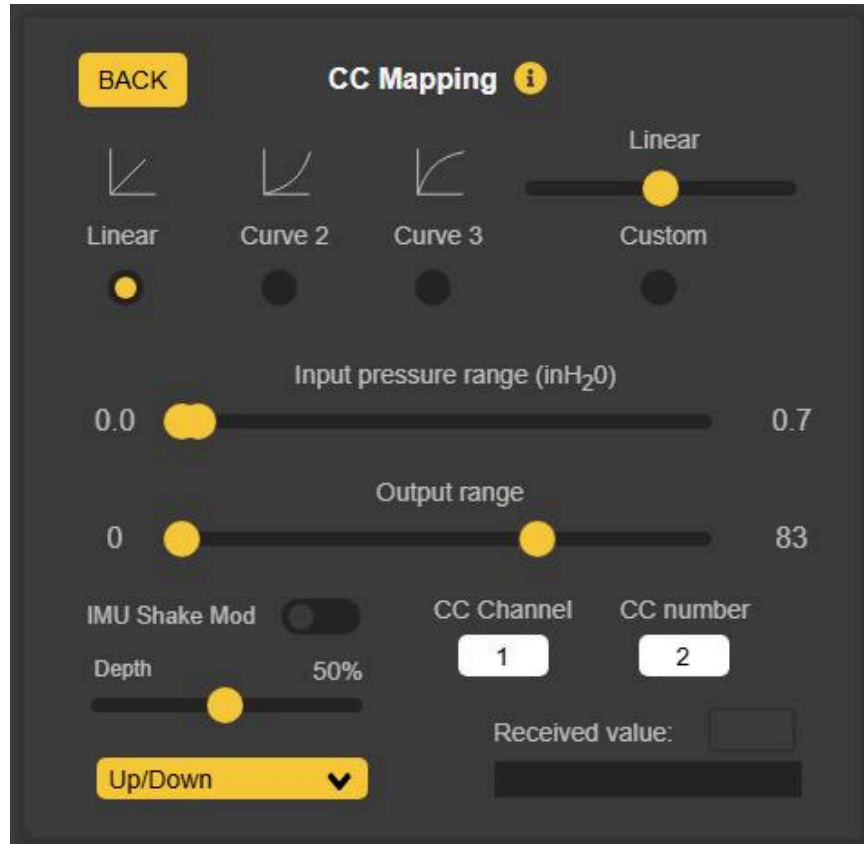
With firmware 4.3 or higher, the pitch expression based on pressure is controlled with two segments:

- A low pressure range, whose min and max input pressure determine a bend along with the low range output bend, from no pressure to the low range max pressure (it will start at the bend amount specified in the low range output bend and end at no bend).
- A high pressure range, whose min and max input pressure determine a bend along with the high range output bend, from the high range min pressure to the high range max pressure (it will start with no bend at the high range min input pressure, and end up with the bend specified in the high range output bend). If the pressure exceeds the high range max input pressure, it will either be clamped at the specified bend amount (if **clamp upper bend** is on), or it will continue with the same bend slope with more pressure.

The low and high ranges will not overlap. Any gap between the low range max, and the high range min will be a stable portion of the pressure range where no pitch bend occurs. You can see exactly how input pressure maps to output bend in cents with the interactive meters for both input pressure and output bend which are behind each pair of sliders.

The **Curve low** and **Curve high** sliders determine the curve for mapping the low and high pressure ranges to pitch bend. The value shown represents the exponent of the power equation used for mapping. Linear mapping will be used if the slider is positioned at "1.00". Double-clicking (or double-tapping) the sliders will return them to the linear setting.

Mapping CC, Velocity, Channel Pressure, and Key Pressure



The input pressure range can be mapped to the output (CC, velocity, channel pressure, or key pressure) range using one of three curves: linear, a power curve (curve 2), or an inverse power curve (curve 3), or (with firmware 4.7 or higher) a custom curve value. Adjusting the sliders below controls which portion of the overall breath/bag pressure is mapped to which portion of the possible MIDI output range of 0-127. When mapping pressure to CC you can choose a CC channel and CC number. Also note that the WARBL₂ will only be sending those messages if the corresponding switch in the main Pressure Mapping panel is

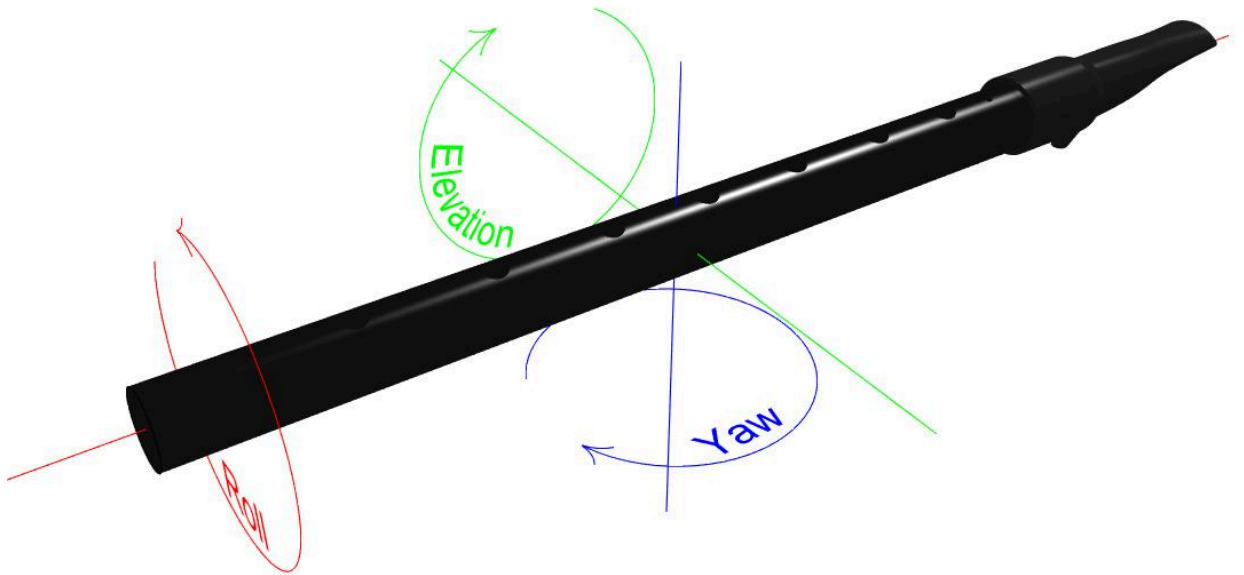
turned on. In the **Channel Pressure** mapping panel, you can turn on the **MPE+** switch to send 14-bit MPE+ specifically for use with Haken Audio hardware.

Beginning with firmware v. 4.3 it is possible to turn on the **IMU Shake Mod** switch to modify the pressure mapping by shaking the WARBL₂ along its long axis. This can be useful, for example, to modify volume by shaking the WARBL₂. It could also be used along with **Shake vibrato** (see the IMU mapping section below) to control both pitch and volume (or another type of expression) by shaking the WARBL₂. You can select the “depth” of the modification as well as the direction (in the drop-down menu below the depth slider).

The **Received value** field and bar at the bottom show the MIDI messages received from the WARBL₂ that correspond to the current panel. For example, if you are mapping breath pressure to CC 7 on channel 1, the display will show only those specific messages. Also, note that the WARBL₂ will only send messages if the corresponding switch in the main panel is turned on.

IMU mapping

The Inertial Measurement Unit (IMU) continuously tracks acceleration and angular rotation in three dimensions. This information is converted to roll, elevation angle (“pitch”), and yaw, which can be then mapped to MIDI CC messages and/or pitch bend.



IMU Mapping ⓘ

Send roll as CC	☐	Map...
Send elevation as CC	☐	Map...
Send yaw as CC	☐	Map...
Send roll as pitch bend	☐	Map...
Send elevation as pitch bend	☐	Map...
Send yaw as pitch bend	☐	Map...
Control register with elevation	☐	Map...
Shake vibrato	☐	Down Only ▼
Shake vibrato depth (percent)		28

A horizontal slider bar is positioned below the 'Shake vibrato depth' label, with a yellow circular knob currently set at the 28% mark.

There is also an option to play pitch bend vibrato by shaking the WARBL₂ along its long axis. You can adjust the amount of **shake vibrato** and also select the “direction” of pitch bend: selecting **up/down** means that the pitch will be altered both above and below the normal pitch for the note, corresponding to the direction of movement of the WARBL₂. **Down/up** has the same effect but opposite the direction of the movement of the device. **Up only** causes the pitch to only be raised above the current note, and **down only** causes the pitch to only be flattened.

To **map roll, elevation, or yaw to CC messages**, turn on the appropriate switch and then click **Map**. A panel similar to the one below will be shown, allowing you to select the input range as well as the range of output CC values. At the bottom there is a readout and bar that will display CC messages received from the WARBL₂ that match the current settings in the panel (note that the switch on the previous panel must also be turned on to cause the WARBL₂ to actually send any messages). For mapping the roll and yaw, there is an option to **center the input range**, which ties the input sliders together to keep the input range centered on zero.

Roll Mapping

Center input range ☒

Input range (degrees)

-90

90

Output range

0

127

OK

CC Channel

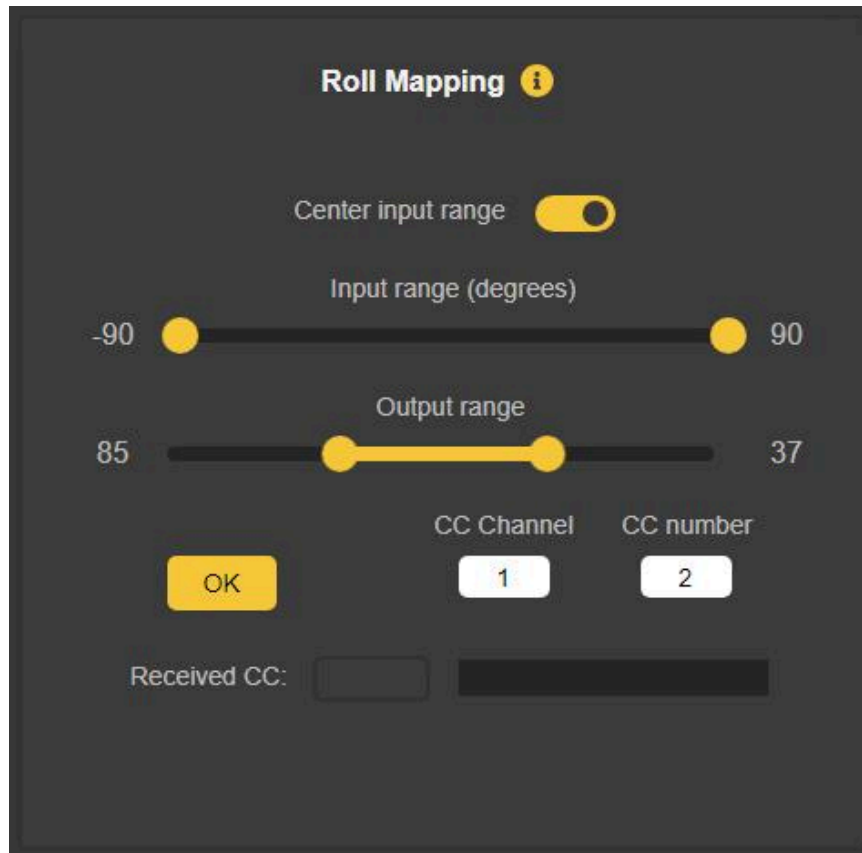
1

CC number

2

Received CC:

Note that it is possible to reverse the output range of the IMU mapping by moving the output sliders past each other as shown below. When the output sliders are reversed that are connected by a yellow bar as shown.



The yaw has a tendency to drift slightly over time, so you may want to occasionally **recenter the yaw** (set it to zero). It is also often desirable to recenter the yaw depending on the rotation of your body. There are three ways to recenter the yaw: There is a button in the WARBL₂ Settings panel to recenter it instantly, you can assign a button action to recenter it (see the Button Behavior panel documentation), or you can use the **auto-center yaw** option in the yaw mapping panel. The auto-center yaw option will reset the yaw to zero when you begin playing after a pause of a certain length of time. You can choose the length of the pause with the **Minimum pause** slider.

Yaw Mapping ?

Center input range ☒

Input range (degrees)

-180 180

Output range

0 127

CC Channel	CC number
1	2

Received CC:

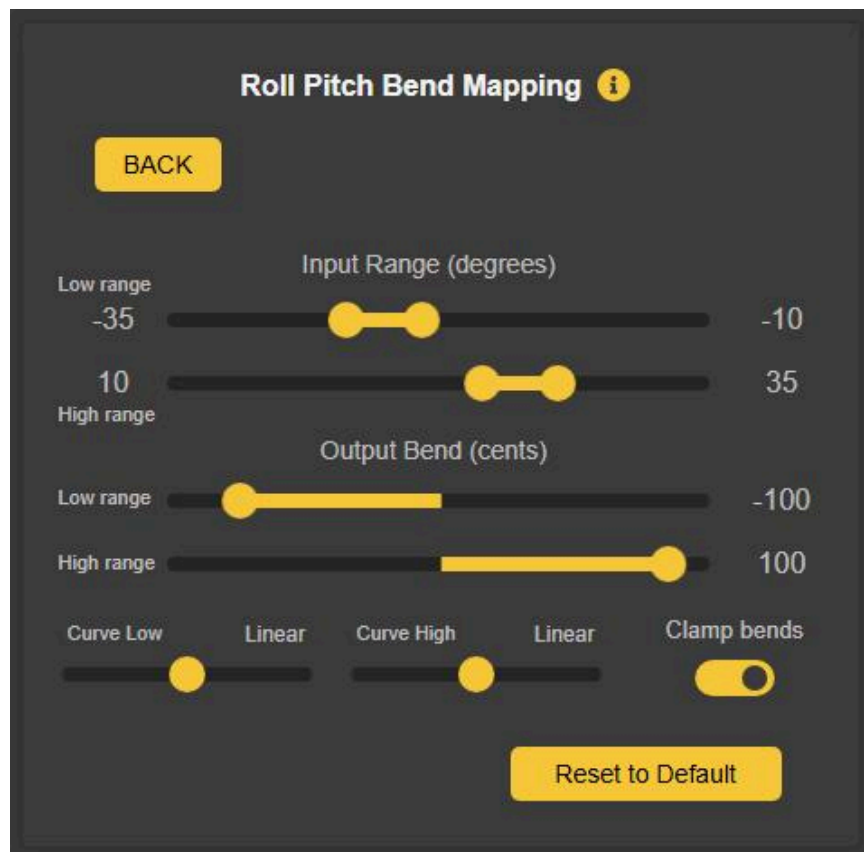
Auto-center yaw ☐

Minimum pause (seconds)

0.25

With firmware 4.4 or higher, IMU rotation can be mapped to MIDI pitch bend messages to “bend” notes by rotating the WARBL₂. To enable this, turn on the appropriate switch in the **IMU Mapping Panel** (for example **Send roll as pitch bend**, and then click the adjacent **Map...** button.

The interface in the IMU > Pitch bend mapping panels (for example the **Roll Pitch Bend Mapping panel** is similar to the one in the **Pitch Expression Pressure Range Override** panel.



The interface consists of two segments that are independently mapped to pitch bend. The segments are:

- A low IMU angle range, whose min and max input angle are mapped to the low range bend output.
- A high IMU angle range, whose min and max input angle are mapped to the high range bend output.

If the angle exceeds the high range max input angle or is less than the low range minimum input angle, it will either be clamped at the specified bend amount (if the **Clamp bends** switch is on), or it will continue to map the same bend slope in the outer angle ranges. Turning off **Clamp bends** can be useful for sending more than one semitone (100 cents) of pitch bend.

The low and high ranges will not overlap. Any gap between the low range max and the high range min will be a stable portion of the IMU angle range where no pitch bend occurs. This is useful for “playing in tune” when no pitch bend is desired. You can use the pitch bend display in the **Finger Vibrato and Slide** panel to see the results of the IMU mapping. The display may concurrently show other pitch bend that the WARBL₂ is sending (for example finger vibrato or IMU shake vibrato) because the pitch bend messages are cumulative.

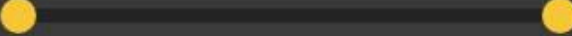
The **Curve low** and **Curve high** sliders determine the curve for mapping the low and high angle ranges to pitch bend. The value shown represents the exponent of the power equation used for mapping. Linear mapping will be used if the slider is positioned at “1.00”. Double-clicking (or double-tapping) the sliders will return them to the linear setting.

There is also an option to control the **register using the IMU elevation angle**. You can choose the total number of registers and the input elevation range. The number of semitones in each register is controlled by the **Register semitones** setting in the **Note Trigger and Register Control** panel. The normal playing range (based on the “transpose” setting) will be at the bottom of the range and

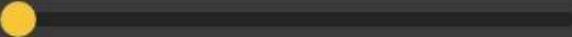
you will be able to play up to four registers above that. If overblowing or thumb register control is also enabled you will be able to use both methods of register control simultaneously.

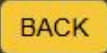
Control Register with Elevation

Input range (degrees)

-90    90

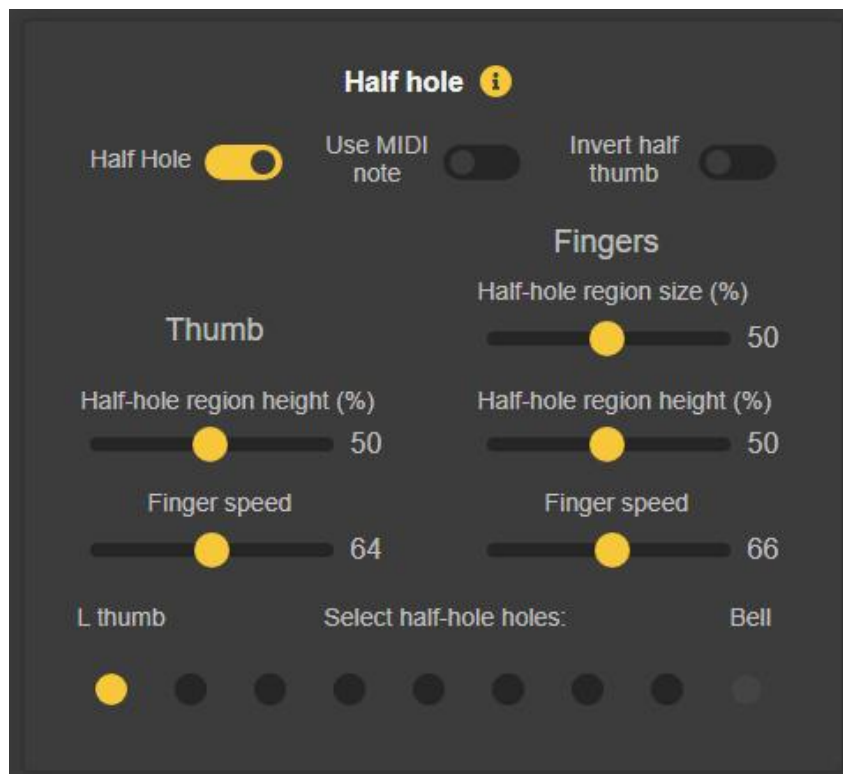
Number of registers

  2



Half Hole

Turning on the **Half hole** switch makes it possible to play semitones or use the thumb to play three registers by covering holes halfway. You can select which finger to use for half holes at the bottom of the panel.



Playing semitones

When using half holes to play semitones, the pitch is normally changed by using MIDI pitch-bend messages. To send new MIDI notes for semitones instead of using pitch bend, turn on the **Use MIDI note** switch. When this switch is turned

on, the **slide** feature will be disabled, because sliding doesn't work as well when many MIDI notes are triggered rapidly.

If using pitch bend to play semitones (with the **Use MIDI note** turned off), half holes can be used with any of the **Finger Vibrato and Slide** modes (any combination of vibrato and slide). When lowering a finger over a hole when using both **slide** and **half hole**, the pitch will gradually bend until the finger is in the half hole region (see below) and then the pitch will be locked to the semitone. Similarly, the pitch will bend if the finger is below the half-hole region.

Semitones can be played by half holing using any hole (except the thumb) that normally causes a pitch change of a full tone (whole step) with your selected fingering chart (including custom charts).

The half-hole region

The **half-hole region height (%)** and **Half-hole region size (%)** sliders allow you to control the size and position of the region above the tone hole where your finger will trigger a semitone. The region height is the height of the center of the region above the tonehole, as a percentage of the full finger-sensing distance. The region size is also a percentage of the overall range. Note that depending on the settings chosen for both parameters, the half-hole region may extend to the top or bottom of the full sensing range (or both). For example, if the half-hole region size is set close to 100% the finger might trigger a semitone anywhere in the sensing range. **Note:** If using MIDI notes to play semitones (the **Use MIDI note** is turned on), the half-hole region always continues down to the closed hole.

The **Finger speed** slider acts as a filter to make it possible to transition from an open hole to a closed hole (or vice versa) quickly without playing a semitone in

between. The speed of the finger is monitored as it is moved through the full finger-sensing range. With the slider moved to the right, a semitone will be triggered while the finger is moving more quickly, making it easier to play semitones. When the slide is moved to the left, the finger needs to be moving more slowly to trigger a semitone, so it is easier to move between whole tones without triggering a semitone in between. The **Finger speed** slide is somewhat similar in functionality to the **Transient filter** but is applied only to half holes. You can use both sliders together to filter out brief half holes as well as other brief notes. **Note:** The **Finger speed** slider has no effect if using **slide** in conjunction with half holes.

Controlling register with the thumb

The thumb can be used to play three registers in the closed, half, and open positions. This is possible with custom charts and existing charts that don't normally use the thumb.

To enable this feature, select **Thumb register** in the **Note Trigger and Register Control** panel. Then, make sure the thumb hole is enabled at the bottom of the **Half Hole** panel.

There is not a **Half-hole region size slider** for the thumb because the half-hole region always extends down to the fully-covered hole (unlike when the fingers are used for sliding and there is a “non-halfhole” region close to the tone hole).

Important: With the **Thumb register** option selected (with or without using the thumb for half holes), if you are using a custom fingering chart, the thumb position in your chart will be ignored and only the first half of the chart will be

used. You do need to have values in the second half of the chart (the part with the thumb hole covered), but those values will be ignored.

You can choose whether each of the three thumb positions (closed, half, and open) plays the first, second, or third register using a combination of the **Invert half thumb** switch (in the **Half Hole panel**) and the **Invert thumb/bell** switch (in the **Note Trigger and Register Control panel**). The combined effect of these switches is shown in the table below:

Invert thumb/bell	Invert half thumb	1st octave	2nd octave	3rd octave
off	off	closed	open	half
on	off	open	closed	half
off	on	closed	half	open
on	on	open	half	closed

Drones Control

The screenshot shows a 'Drones Control' interface with the following elements:

- Title:** Drones Control with an information icon.
- Drone on command:** A dropdown menu set to 'Note on'.
- Channel (1-16):** A numeric input field set to 1.
- Byte 2 (0-127):** A numeric input field set to 51.
- Byte 3 (0-127):** A numeric input field set to 36.
- Drone off command:** A dropdown menu set to 'Note on'.
- Channel (1-16):** A numeric input field set to 1.
- Byte 2 (0-127):** A numeric input field set to 51.
- Byte 3 (0-127):** A numeric input field set to 36.
- Control Methods:** Three radio buttons: 'No drone control' (selected), 'Use "secret" button', and 'Turn on/off with chanter'.
- Use pressure:** A radio button that is currently unselected.
- Learn:** A yellow button.
- inH₂O:** A numeric input field set to 0.

WARBL₂ makes it possible to control drones in MIDI bagpipe apps by assigning commands for turning the drones on and off, and then choosing from a few methods for sending those commands. You can choose to send either Note On, Note Off, or CC messages, assigning the channel, byte 2, and byte 3. For Note On and Note Off messages, byte 2 is a MIDI note number and byte 3 is velocity. For CC messages, byte 2 is the CC number and byte 3 is the value.

The commands used for turning drones on and off will depend on the MIDI host app. For example, the Celtic Sounds app uses a Note On command both for

turning drones on and turning them off. In some apps such as Universal Piper, you can choose which command you'll use to control the drones.

Choosing **Use secret button** will allow you to turn the drones on and off by pressing button 1 while covering the right-hand index-finger hole.

Choosing **Turn on/off with chanter** will cause the drones to be on whenever the chanter is playing. This will be true no matter how you control the chanter, so will work in **bagless mode** as well as using the pressure sensor.

Selecting **Pressure** will turn the drones on and off at a given pressure threshold, so, for example, you can mimic real bagpipes by having the drones come on at a baseline pressure and the chanter come on at a higher pressure. You can either **"learn"** the pressure or enter it manually, as with the note trigger pressure.

It is also possible to assign any button action to turn the drones on and off. To do this, leave the option in this panel set to **No drone control** but then assign a button action to **Turn drones on/off** in the Button Behavior panel (see below).

Button Behavior

Momentary				Channel (1-16)	Byte 2 (0-127)	Byte 3 (0-127)
☐	Click 1	none		0	0	0
☐	Click 2	none				
☐	Click 3	none				
	Hold 2, click 1	none				
	Hold 2, click 3	none				
	Long-press 1	none				
	Long-press 2	none				
	Long-press 3	Power down				
	Sip mouthpiece	none				
	Shake	none				

The three buttons can be programmed to perform different actions. This is useful for sending special commands to MIDI apps or customizing the ability to switch modes. **Note** that it can be confusing if you assign different buttons actions in the three different presets, so it may be useful to use the same actions in all three.

New to WARBL₂ there are two additional actions included with the “button” actions: You can either “**sip**” the mouthpiece or **shake** the device to trigger commands. Because these are both relatively easy to do accidentally, it is recommended to only use these for commands that won’t have adverse effects or cause confusion if triggered accidentally. Example uses might be recentering the yaw or MIDI panic.

If a button click is configured to send MIDI Note On/Note Off messages, the normal behavior is to alternate between sending Note On and Note Off

messages each time the button is "clicked" (the action actually occurs when the button is released). There is also the option to choose "**momentary**" behavior, meaning that the button will send a Note On message when pressed and a Note Off message when released. This is useful if you only want a note to be on when the button is held down, for example for mimicking uilleann pipe regulators. Note: If **momentary** is selected, other actions using that same button will be disabled, because it's not possible to differentiate other intentions from a momentary press.

New in firmware version 4.2, you can turn on the **Use button double-clicks** switch to require double-clicks instead of single-clicks for all three buttons. This can help prevent accidentally clicking the buttons. If a **Momentary** switch is turned on for a particular button, then the **Use button double-clicks** switch is ignored for that button.

If you are sending Note On/Note Off messages, **byte 2** is the MIDI note number, and **byte 3** is velocity. If you are sending CC messages, **byte 2** is the CC number and **byte 3** is the value. Please note that CC messages 102-119 on channel 7 are reserved for the Configuration Tool, so WARBL cannot be programmed to send additional messages in this range.

If you are sending a CC message and the **momentary** switch is turned on, a CC message with a value of 0 will be sent when the button is released. When initially clicked, a message with the value that you entered for **Byte 3** will be sent. This behavior can be used for on/off CC messages like CC 64-69.

Selecting **Send PC** will send a Program Change message to the selected channel. This would be useful, for example, to select a particular preset in a MIDI host app. **Increase PC** and **Decrease PC** are different in that they assume a Program Change value of 1 when WARBL₂ is started, and then increase or decrease it from there. For example, clicking a button assigned to **Increase PC** after starting WARBL₂ will send a PC message of 2 on the selected channel.

These commands are useful for consecutively stepping through the available sounds in a MIDI app.

The **momentary** switches can also be used when you have selected **Octave shift up**, **Octave shift down**, **Semitone shift up**, or **semitone shift down**. This allows you to use a button to momentarily shift the octave or key, extending the range of the instrument and allowing any button to act as a "key" for sharpening or flattening a note.

A button can be assigned to **begin autocalibration**. This can be useful if you anticipate needing to autocalibrate when you don't have access to the Configuration Tool. However, you'll want to use a button combination that can't be triggered accidentally, for example holding button 2 and clicking either button 1 or button 3. Also, when you begin autocalibration with a button combination, you need to be careful not to cover any of the sensors while pressing the buttons, because if you are pressing firmly on the sensors when autocalibration begins it will give WARBL₂ incorrect examples of covered sensors.

By default, a long-press of button 3 will **power down** the WARBL₂. If you'd prefer not to use a button to power down the device you can choose to let it power down automatically (see the WARBL₂ Settings section).

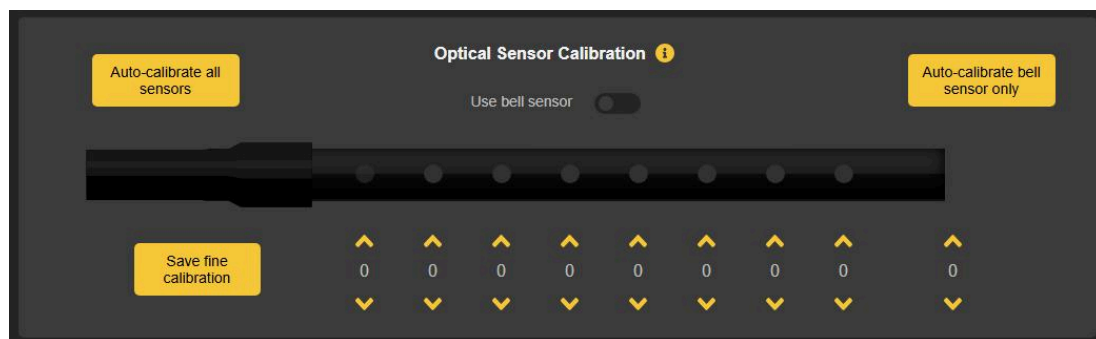
On WARBL₂ you can also set a button action to **recenter the yaw**, or **show the battery level**. When triggered, the latter will blink the LED in a turquoise color from 1-10 times according to the battery level percentage. If the battery level is lower than 10%, the LED will blink once in red.

The **Register hold** option (WARBL₂) allows you to temporarily prevent the register from changing by overblowing. This can be useful for changing the dynamics in the current register without accidentally changing the register.

The **Toggle slide mode** option switches back and forth between the **Slide and vibrato mode** and the **Legato slide/vibrato** mode, assuming either of those was previously selected. This can be used with or without the momentary switch turned on if assigned to a button click.

The **"secret"** button commands are a few additional hard-coded actions that involve covering certain tone holes while clicking button 1. These can be useful if normal button actions are used for other functions. They give you an alternative way of changing slide/vibrato mode and preset. With the "secret" commands turned on, covering only hole L3 while pressing button 1 will change the slide/vibrato mode, and covering only R3 while pressing button 1 will change the preset. Note that the **Use "secret" button commands** switch does not have to be on to use the "secret" drone control button, if that option is selected in the Drone Control panel.

Optical Sensor Calibration



The optical sensors may occasionally need to be calibrated, for example when the device is new, if a new person is playing the device, or the sensors are a bit dirty. Calibration can also be used in a pinch to compensate for having dirty hands, but it's easier to wash your hands...

Note that there is also a video about calibrating the sensors on the WARBL web site.

If the calibration is far off or you want to calibrate quickly, you can run **auto-calibration**. Begin by holding WARBL₂ in a well-lit room (but not direct sunlight) in such a way that no tone holes are covered (important because it gives WARBL₂ examples of raw sensor readings) and then click **Begin auto-calibration**. The LED on WARBL₂ will turn on and then you'll have ten seconds to lightly place your fingers over all the tone holes. Pressing harder on each tone hole will make it more difficult to "cover" after auto-calibration. After this the calibration should be reasonably close. If you plan to use the bell sensor, rest it very gently on whatever surface you intend to use with it while also covering the tone holes with your fingers (or you can calibrate it separately as described below). Note that auto-calibration is always saved immediately; it is not necessary to click **Save fine calibration** afterward.

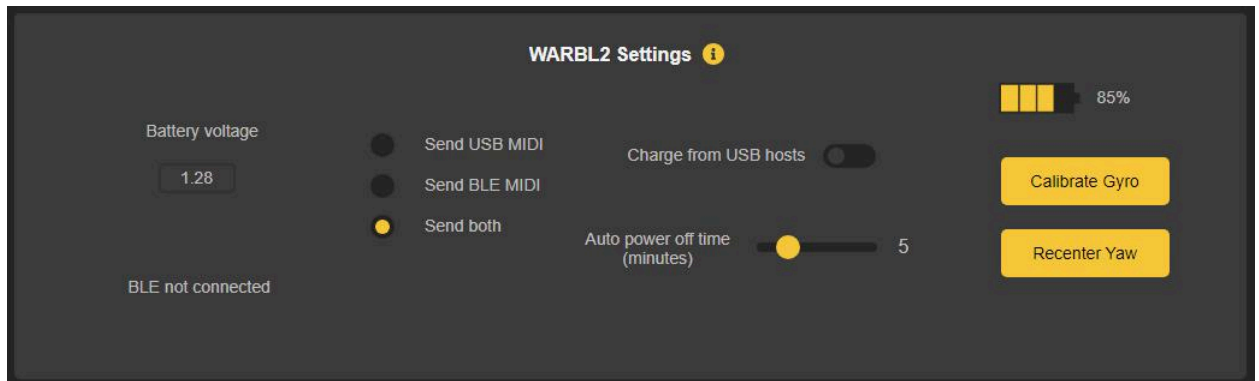
You also have the option of auto-calibrating only the bell sensor. This is useful because the sensor may respond differently depending on the type of clothing you're wearing. To do this, hold WARBL up in the air so the sensor is completely uncovered, click **Auto-calibrate bell sensor only**, and then lower the sensor very lightly onto your knee. After five seconds, the LED will turn off and the sensor will be calibrated.

There is a switch to **Use the bell sensor**. This is turned off by default but you can turn it on if you want to be able to use the bell sensor to stop all sound when you cover it along with all the other tone holes. If you have selected one of the uilleann pipes fingering charts, those will automatically use the bell sensor, and you don't need to turn the switch on. You also don't need to turn it on for calibrating. If you do not need the bell sensor, keeping the switch off will increase battery life by around 10%.

After auto-calibration or for more frequent adjustments, you can just fine-tune each sensor individually. Clicking the up arrow will make it easier to "cover" the tone hole, while clicking the down arrow will make it more difficult. Note that when you plug in WARBL₂, values for each sensor will always read zero, and any changes will be relative to the current calibration. Changes to individual sensors will take place immediately so you can test them as you are changing them, but you must click **"Save fine calibration"** for the changes to be saved. Otherwise the calibration will be reset when WARBL₂ is unplugged.

WARBL2 Settings

This panel contains information and settings specific to WARBL₂. Changes to this panel are saved immediately; it is not necessary to "save settings for the current preset", and as with sensor calibration, these settings apply to the entire device rather than one specific preset.



The **battery voltage** is shown as a convenience, but the **percentage monitor** on the far right is usually a more reliable indicator of charge level. As mentioned previously, it is an estimate and becomes more accurate after letting the battery run all the way down (until the WARBL₂ powers off) and then charging fully. When charging, a charging icon is shown next to the percentage monitor. If the charging icon is blinking, it may be an indicator of a battery fault such as very low or high temperatures or a battery that is too old to charge.

The BLE status is shown in the bottom left. When connected by BLE it will show the connection interval in milliseconds.

If the WARBL₂ is connected to both BLE and USB, you can choose whether to send MIDI data over either or both connections. If only one connection is available, that connection will be used regardless of this setting. The WARBL₂ will also communicate with the Configuration Tool by all available connections regardless of this setting.

The WARBL₂ can be charged from any USB host (for example PC, Mac, tablet, or phone) or from a USB charging brick. Charging from a USB host can be turned off, which is recommended if you are using USB MIDI and the WARBL₂ is often connected to a host. This will help save the battery of your host device.

To save battery life, the WARBL₂ will shut down automatically after some time if no MIDI notes have been sent or commands have been received from the Configuration Tool. You can adjust the power off time with the slider. Moving the slider to its maximum will prevent the WARBL₂ from ever shutting off automatically. An infinity symbol (∞) will be shown. This setting is generally not recommended because it can cause the battery to be depleted. You can also assign a button action to power off the WARBL₂ in the Button Behavior panel.

The gyroscope is calibrated when the WARBL₂ is made, but you may wish to calibrate it again occasionally (especially if the roll and elevation seem to drift slightly when the WARBL₂ is stationary). To do this, set the WARBL₂ on a stationary surface and click **Calibrate Gyro**. If you are using IMU yaw to send MIDI data, you can recenter the yaw by clicking the button. You can also assign a button action to recentering the yaw in the Button Behavior panel, and there is also an option in the Yaw Mapping panel to automatically recenter the yaw when there is a break in notes.

8. Using bagpipe bags and the integrated bell sensor

The optional bagpipe kit includes a simple collapsible plastic bottle that can be used as a bagpipe bag. Simply inflate the bag to the desired amount. Then plug the tube into the rubber stopper in place of the mouthpiece tube. Please see the Configuration Tool section above for information on settings for controlling the register, “learning” calibration pressure, and drones control options when using a bag.

It is also possible to use real bagpipe bags, though the method of attachment will vary. Please feel free to contact me for ideas about connecting other bags.

The integrated bell sensor is designed to allow closed piping, as with uilleann pipes. It is similar to the other tone-hole sensors, but is intended to sense when the instrument is “on the knee” or “off the knee”. If the bell sensor is turned on in the Configuration Tool, no note will be sounded if all the tone holes and the bell sensor are covered (with any fingering chart). This simulates an uilleann chanter that is completely closed off. The sensor can also be used for vibrato.

9. Care and maintenance

It is best to store the WARBL₂ at around room temperature. As mentioned previously, ensure that the power button (button 3) isn’t continuously pressed during storage, for example by a tight-fitting storage container. This will cause the device to stay powered on and will drain the battery. When storing the WARBL₂

for long periods of time it is a good idea to recharge the battery every few months.

The only maintenance required is occasional cleaning. A soft cotton cloth (dampened with water if necessary) can be used for cleaning the outside of the enclosure, and a cotton swab works well for cleaning the toneholes and optical sensors. You can dampen the swab with water and if necessary. A compressed air can like those used for cleaning computers can be used to gently blow any dust out of the interior. You can blow air into the tone holes along the sides of the sensors.

Note: Do not use alcohol or other solvents to clean any part of the WARBL₂. Solvents can damage the acrylic enclosure.

The stopper and mouthpiece can be cleaned with soap and water, or can be boiled to sterilize.

Please remember that WARBL₂ is a sensitive electronic device, and shouldn't be subjected to extremes of heat, moisture, or vibration. If you treat the mechanical parts such as the buttons and USB connector with respect, WARBL₂ should give you many years of service.

10. Updating the firmware

There will occasionally be firmware updates for WARBL₂. When you are connected to the Configuration Tool, it will tell you in the upper right corner if your software is up to date or if there is a new version available. The newest version of the firmware is always available on the [Documentation](#) page. Please feel free

to email info@warbl.xyz or ask on the WARBL Forum if you have any trouble with the installation process.

The easiest way to update the firmware is to use a Windows PC or Mac. Begin by downloading the latest firmware from the [Documentation](#) page.

WARBL₂ has the ability to appear as a USB (“thumb”) drive on your computer. To force it to enter this mode, hold down button 1 (topmost button) on the WARBL₂ while you plug it in. It should appear as a drive called WARBLBOOT. Another way to force it to appear as a drive if it is already plugged in is to double-click the reset button with a toothpick.

Once the WARBL₂ appears as a drive, simply drag the firmware update file onto the WARBLBOOT drive (or copy/paste the firmware). The firmware should update in a few seconds and the WARBLBOOT drive will disappear. At this point the WARBL₂ will have turned back into a MIDI device. The next time you connect to the WARBL Configuration Tool, it should show the new firmware version in the top right corner.

Note: Often both Windows and Mac will give an error message when the WARBL₂ turns back into a MIDI device because the computer is not expecting the “drive” to disappear. This is okay and the update is still successful. At the point when the error appears the update is already complete, so it doesn’t matter which option you click to dismiss the error.

It is also possible to perform the update process on mobile devices if the WARBL is connected with a USB cable.

11. LED Indication

The LED on the WARBL₂ can produce any color, and while the LED is mentioned in other sections of this manual, the meaning of the various colored indications are summarized below:

Green:

- Blinks once when powering on
- Blinks once when connecting to or disconnecting from the Config Tool
- Blinks once when simple commands are received from the Config Tool or simple controls are changed with button behaviors
- Blinks three times when saving settings, applying the currents settings to all three presets, or restoring factory settings
- When certain settings are changed, it flashes multiple times to indicate the "number" of the new setting. These include: preset, vibrato/slide mode, register control mode, and octave shift up and down.
- Stays on for 10 seconds when calibrating sensors (5 seconds when calibrating the bell sensor alone)

Blue:

- Blinks quickly when in programming (UF2) mode
- Blinks twice when connecting by BLE
- May blink once when opening or closing a BLE app on certain systems

Red:

- Pulses slowly when charging
- May blink approximately once per second if there is a battery error while attempting to charge (temperature too high/low, battery missing, etc.)

- Lights for 1/2 second when powering down by a button command
- Lights for 5 seconds before shutting down because of low battery level

Teal:

- Flashes to indicate the battery level when that option is triggered by a button action

12. Replacing the battery

The battery should last for several years. If the playing time per charge seems to be getting unacceptably short you can replace the battery yourself. It is a specialty battery but is available in most countries or it can be ordered from the WARBL website. The battery is a 1.2 V, 400 mAH $\frac{2}{3}$ AAA NiMH cell. It can be either a button top or flat top battery. Good-quality brands for this battery include PKCell, Kastar, Dantona, and GP.

To replace the battery, first determine if you have the “old”-style end cap, or the “new”-style cap (made later than January of 2025), based on the photo below. The old cap has a friction fit, and the new cap rotates a small amount to lock in place.



If you have the “new”-style end cap (otherwise scroll down to the next section):

Begin by using a coin or other thin object like the back of a knife blade to rotate the cap counterclockwise slightly until it stops turning (being careful not to mar the end cap).



At this point the battery spring may push the cap partway out. If not, use a thin blade like a razor blade under the edge of the cap to help loosen it and lift it out.

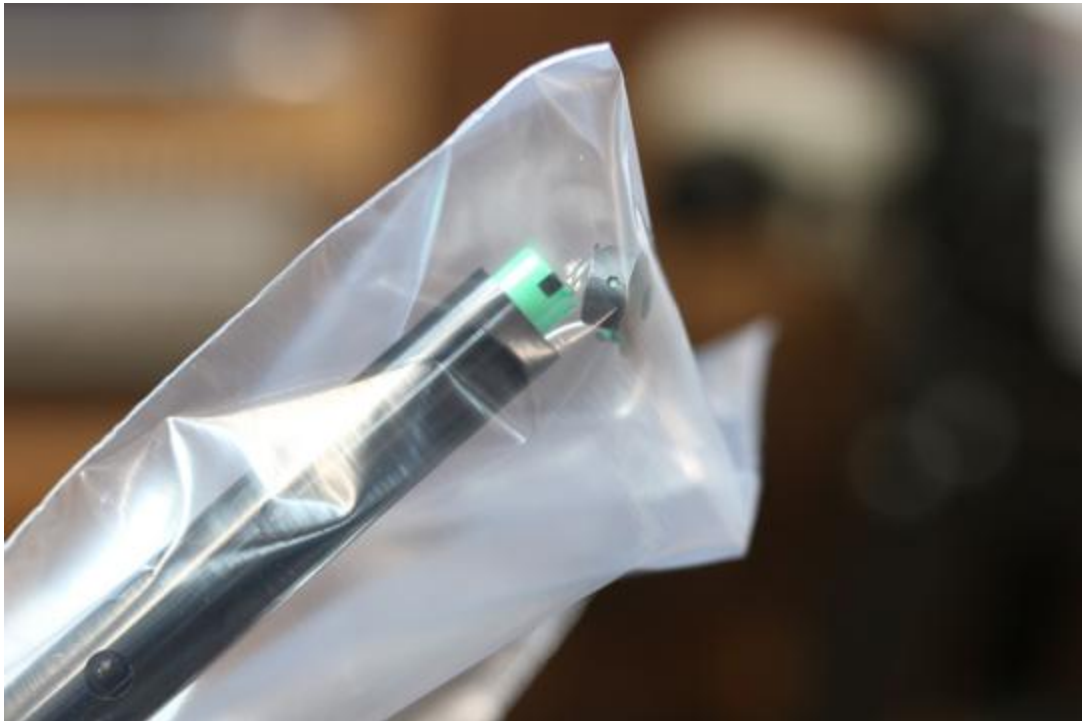


Once you can grab it with your fingers, carefully wiggle the cap out of the tube.

Note: The cap is connected by a cable, so be gentle and don't pull the cap too hard. Once the cap is free you can pull it straight out approximately 1 cm ($\frac{1}{2}$ inch). The cable is extendable and it is okay to pull it this far out of the tube. This will give you space to remove the battery.



The battery is held in place by a small spring-like extension on the circuit board that keeps it from rattling. Because of this, the battery won't simply fall out. You may be able to pull the battery out with tweezers or a strong magnet, but the easiest way to remove it is to place the WARBL₂ in a plastic or cloth bag and “flick” it so that centrifugal force will eject the battery as shown below.



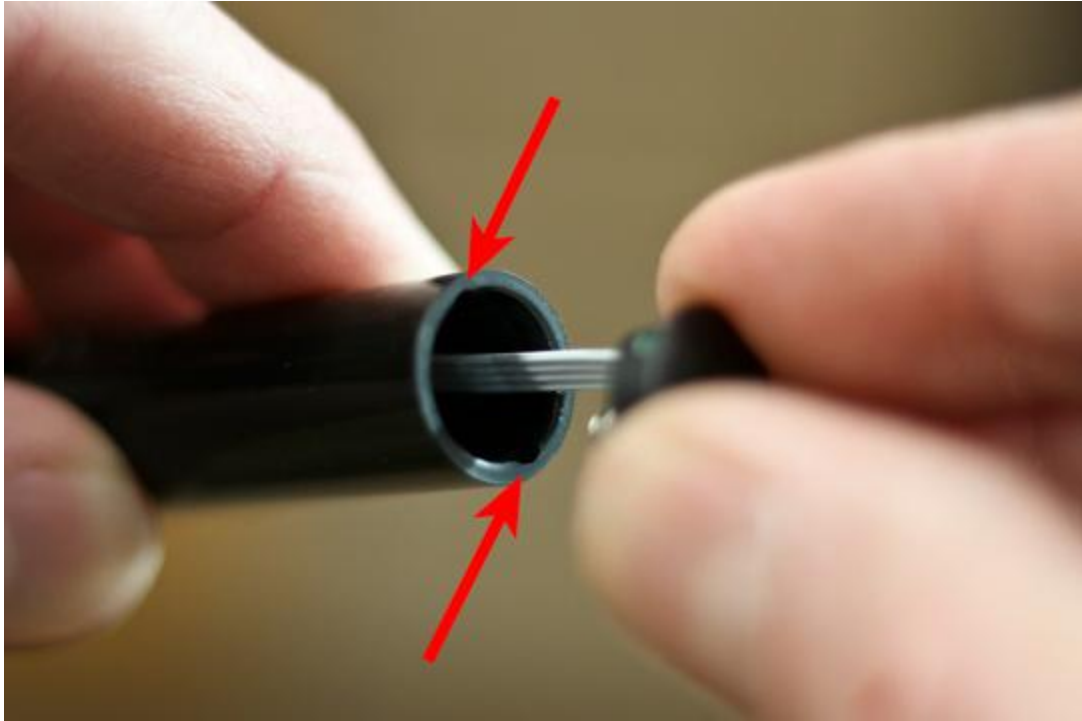
Now you can grab the battery and remove it the rest of the way.

Then insert the **positive end** of the new battery (**Important! The positive end will have a “+” symbol.**) into the tube and push it in with a non-metallic tool (like a pencil) until it stops (you will feel some resistance as you push it past the spring).

Once the battery is fully seated, begin pushing the end cap back in place, inserting the side of the cap away from the cable first as shown below. This will help keep the cable straight as you insert the cap. Try to keep the cable as straight as possible, because if it becomes kinked it will be more difficult to push back inside the enclosure.



Notice that there are two slots inside of the enclosure tube:



There are small green tabs on the side of the end cap that fit into the slots:



If necessary, rotate the cap slightly until the tabs engage in the slots and you can push the cap the rest of the way in place. Then use a coin or other tool to rotate the cap slightly clockwise to lock it in place. Don't try to overtighten the cap—it will only require a small rotation to stay in place.

The battery percentage monitor won't show an accurate percentage until the battery has undergone a full charge/drain cycle. For this reason it is recommended to fully charge the battery (until the LED turns green) after replacing, and then use the WARBL₂ until the battery is completely drained (until it powers down because of a low battery). After this the percentage monitor should reflect the increased capacity of the new battery.

If you have the “old”-style end cap:

Begin by removing the end cap. The cap is connected by a cable and contains a circuit board with the negative battery terminal and the bell sensor. Use a single-edged razor blade or other sharp blade to loosen the cap. Insert the blade between the edge of the cap and the WARBL enclosure (tube) as shown below. It will only go in about 1 mm.



Next, continuing to hold the blade perpendicular to the tube, use your thumb to push the blade and the edge of the cap upward. Don't try to pry with the blade, as this can damage the end cap. Simply use the blade to push the cap out of the

tube. If necessary, rotate the tube, reinsert the blade on the other side of the cap, and continue to push the cap upward.

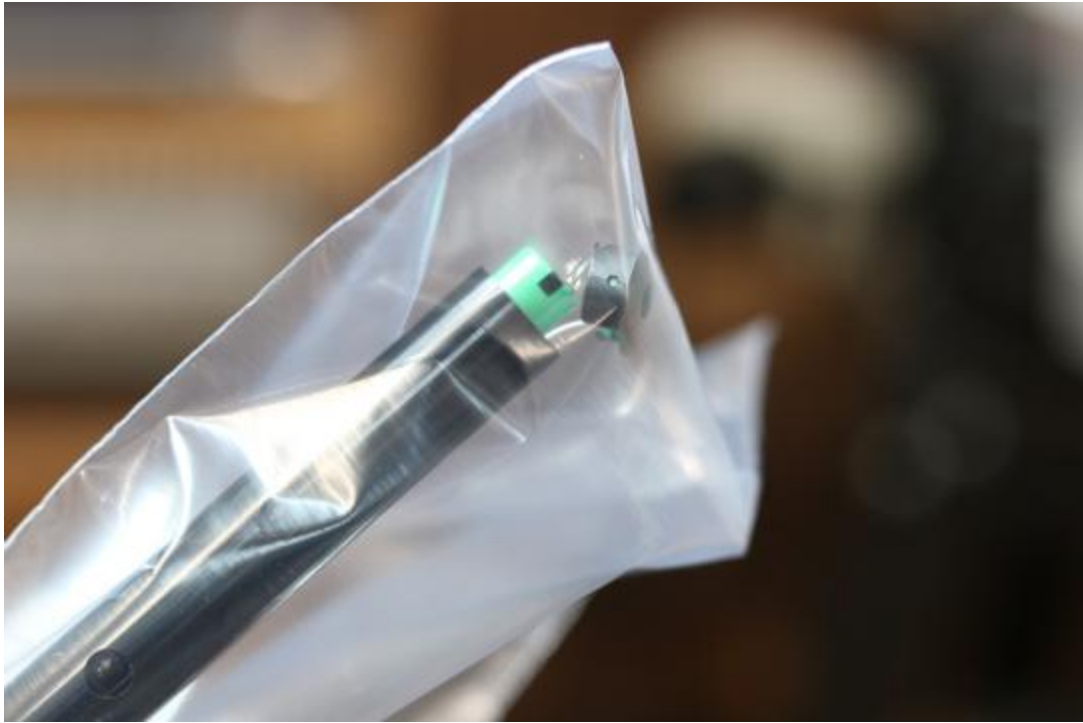


Once you can grab it with your fingers, carefully wiggle the cap out of the tube.

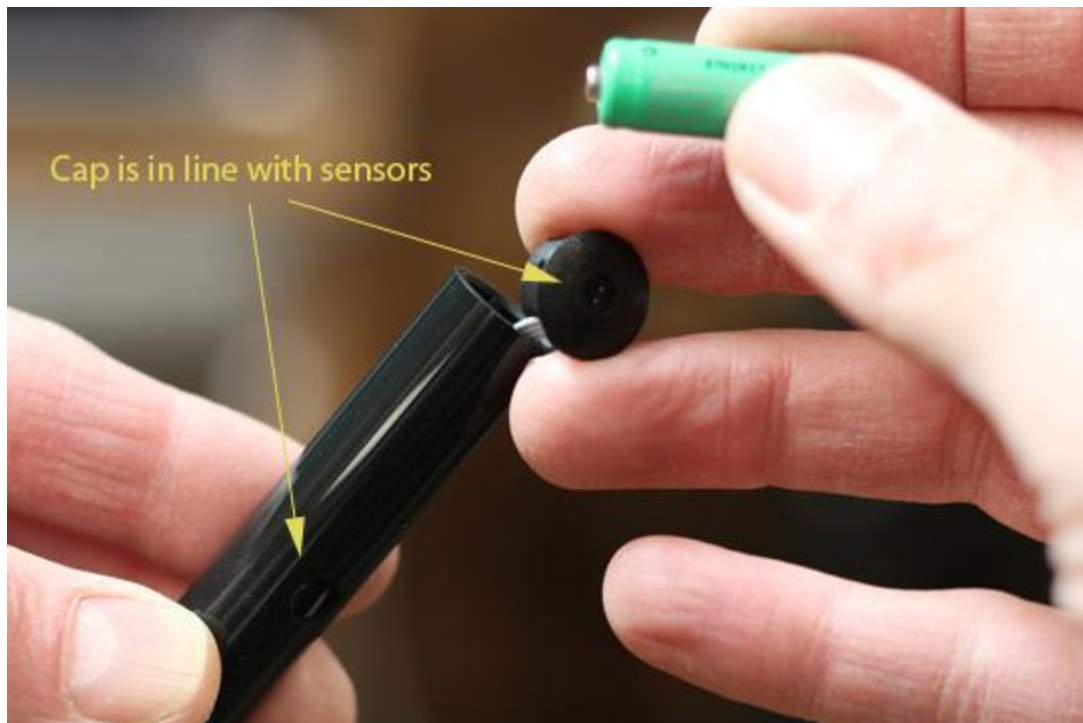
Note: The cap is connected by a cable, so be gentle and don't pull the cap too hard. Once the cap is free you can pull it straight out approximately 1 cm ($\frac{1}{2}$ inch). The cable is extendable and it is okay to pull it this far out of the tube. This will give you space to remove the battery.



The battery is held in place by a small spring-like extension on the circuit board that keeps it from rattling. Because of this, the battery won't simply fall out. You may be able to pull the battery out with tweezers or a strong magnet, but the easiest way to remove it is to place the WARBL₂ in a plastic or cloth bag and “flick” it so that centrifugal force will eject the battery as shown below. Now you can grab the battery and remove it the rest of the way.



To install the new battery, rotate the end cap slightly towards the top side of the WARBL₂ (the side with the tone hole sensors) as shown below. This will prevent the cable from getting pinched when you insert the battery.



Now insert the **positive end** of the battery (**Important! The positive end will have a “+” symbol.**) into the tube and push it in with a non-metallic tool (like a pencil) until it stops (you will feel some resistance as you push it past the spring).



Once the battery is fully seated, begin pushing the end cap back in place, inserting the side of the cap away from the cable first as shown below. This will help keep the cable straight as you insert the cap. Try to keep the cable as straight as possible, because if it becomes kinked it will be more difficult to push back inside the enclosure.



Once the first side of the cap is in place, insert the other side and push the cap in. If you'd like you can test the WARBL₂ when the cap is halfway seated. If it doesn't work, there's a chance that the cable has become bent and is stuck between the battery and the spring contact. If this is the case, remove the cap, straighten the cable, and install the cap again. After testing the WARBL₂, push the cap in until fully seated.



The battery percentage monitor won't show an accurate percentage until the battery has undergone a full charge/drain cycle. For this reason it is recommended to fully charge the battery (until the LED turns green) after replacing, and then use the WARBL₂ until the battery is completely drained (until it powers down because of a low battery). After this the percentage monitor should reflect the increased capacity of the new battery.

13. Troubleshooting

Please see the WARBL [FAQ page](#) for troubleshooting. You can also ask questions on the [WARBL Forum](#) (or search for existing questions), or email info@warbl.xyz.

Warranty

WARBL is guaranteed for a period of one year from the date of purchase. The warranty does not cover damage to mistreatment or misuse. The customer is responsible for the cost of shipping for warranty repairs. If your WARBL is damaged or malfunctioning and it is beyond the warranty period, please still contact me as I am usually able to make accommodations. Contact for warranty service: info@warbl.xyz.

Return Policy

I am happy to accept returns within 30 days of delivery. I'll also evaluate other returns on a case-by-case basis. Please contact me at info@warbl.xyz.

Technical Data

Dimensions: 8.9" X 0.625"

Weight: 1.2 oz

Operating frequency: 64 MHz

Operating voltage: 3.0 V

Typical power consumption: 30 mW, 500 mW (100 mA @ 5 V) when charging

Interface: USB 2.0 full speed, BLE MIDI 5.3

This product complies with FCC Part 15 Class B and also with CE Directives EN 55032, EN 55103-2, and 2015/863, and carries the CE mark accordingly.

Contains FCC ID: 2AAQS-ISP1807

This device complies with part 15 of the FCC rules. Operation is subject to the following two conditions. (1) This device may not cause harmful interference (2) This device must accept any interference received, including interference that may cause undesired operation. Caution: Any Changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

WARBL is Open Hardware and is licensed under the CERN OHL version 1.2 and the GNU General Public License.

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